
‘Focus Mode’ For Digital Wellbeing - A Cognitively Informed Design For A Smartphone Launcher

A thesis submitted in fulfillment of the requirements

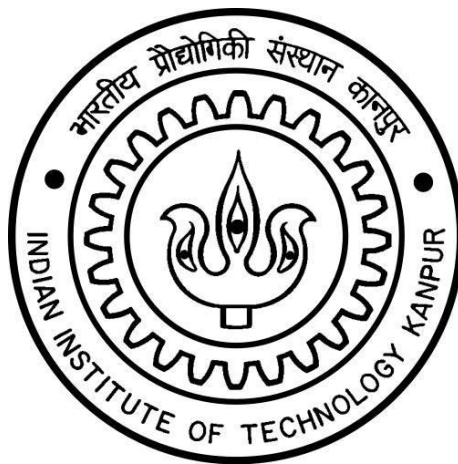
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Master of Science (by Research)

by

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January 2022



Certificate

It is certified that the work contained in this thesis entitled “*‘Focus Mode’ For Digital Wellbeing - A Cognitively Informed Design For A Smartphone Launcher*” by *Sruzan Lolla* has been carried out under my supervision and that it has not been submitted elsewhere for a degree.

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Declaration

This is to certify that the thesis titled “*‘Focus Mode’ For Digital Wellbeing - A Cognitively Informed Design For A Smartphone Launcher*” has been authored by me. It presents the research conducted by me under the supervision of Dr. Harish Chandra Karnick. To the best of my knowledge, it is an original work, both in terms of research content and narrative, and has not been submitted elsewhere, in part or in full, for a degree. Further, due credit has been attributed to the relevant state-of-the-art and collaborations (if any) with appropriate citations and acknowledgments, in line with established norms and practices.



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Abstract

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Thesis title: **‘Focus Mode’ For Digital Wellbeing - A Cognitively Informed Design For A Smartphone Launcher**

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It is possible to design digital spaces that exploit users while the users are unaware of how they are being used to benefit someone else. There are abundant digital means for almost every aspect of human life. To stay in the market, businesses are in the game of seeking continued user attention. In order to take control and leverage the capabilities of technology, it is important to design technology as per the needs of the human mind and resist any exploitation of its limitations. Digital Wellbeing envisions an optimal balance of technology usage to promote work productivity among other goals to achieving wellbeing.

A number of design choices make up today’s ubiquitous UI on smartphones. Traditional smartphone interfaces propagate interruption-driven usage there by hindering task-driven usage during smartphone pick up interaction. We hypothesise that Interruptions and Distractions both lead to Goal Slippage. Our study focuses on the pervasive menu-grid interaction for smartphone app selection and challenges the heuristic design principle ‘Recognition rather than Recall’ that recommends it against plain search. We investigate the phenomenon of Goal Slippage in digital devices through an experiment that captures the

extent of Goal Slippage and Loss of Content with the altering presence of external interruptions and distractions. We find that external interruptions have a significant effect on response times suggesting Goal Slippage but distractions do not have any effect. However, a wider spread of response time is established during distractions suggesting disruption if not complete goal slippage.

Based on the results, we attempt to design a smartphone launch interface ‘Focus Mode’ for digital wellbeing backed by a set of design principles.

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Chapter 1

Introduction

Man-made spaces can be molded into a shape that the creator desires. The spaces can be shaped in order to satisfy the specific intention(s) of the creator. Often the intentions behind the design are not visible at the surface level of a space. Even when there are hints of what the intentions may be, it might take a skilled professional from the same domain to decode and understand them. This makes it possible to design spaces that exploit users while the users are unaware of how they are being used to benefit someone else.

In the creation of physical space, one such extreme can be seen in the design of Casino spaces. Casinos are deliberately designed to promote addiction and exploitation of the players[41]. When visitors enter a casino, they might lose the sense of time and get dissociated from the outer world within minutes. An engaged player can even get dissociated from their own self while caught up in the never-ending play-loop. Several deliberate design choices go into the design of such spaces. Curving hallways welcome the user and guide them through a predefined path. It deliberately avoids right-angle intersections, which require momentary pauses that may involve decision-making. A low ceiling ensures that the visitor is in a secured zone and provides a sense of safe shelter. Sound-damped wall coatings reduce the overall sound level of the space and thereby pass only the high amplitude tones and coin buzzes. These and many such choices facilitate the creation of a dissociated experience for the player. This can be compared with the design of spaces with the intention of reflection in thought and self-awareness, such as religious buildings. The high ceiling and ample ventilation make visitors connect with their thoughts and reflect upon them. A similar observation of extremes can be extended to digital spaces designs.

1.1 Digital Design - The Problem

Gadgets and mobile devices are at the center of the current world. There are abundant digital means for almost every aspect of human life. In the game of seeking attention, digital designs are centered on user experience and user affection. An interface has to seduce and convince users within seconds. In order to make users use it repeatedly, an emotional connection has to be established. The radical shift in design intention from usability to emotional engagement has shaped gadgets and digital applications and engineered their usage.

In an attempt, the author[22] frames the situation as a multi-faceted problem where the stakeholders are in a double bind. There are three stakeholders in digital space design, namely End users, Designers, and the Companies behind the digital gadgets/applications. Each stakeholder has succumbed to the current situation for their own reasons and perspectives. End users can find themselves using gadgets longer than they intend to. They can sense the irresistible urge to use the device many times throughout the day. However, end-users are not in a position to verbalize their problem and instead blame themselves for their usage behavior. Due to the forced dependency on gadgets in today's world, they can not afford to do away with their gadgets completely. Designers who create digital designs serve their companies and have no capacity to argue against the persuasive design ideology. They have to reinforce the same design paradigm and consider persuasive design as a best practice. Finally, to stay relevant in the market, the Companies have to label self-affection as user affection. Else, they might soon get replaced by their competitors and disappear from the market. Thus each stakeholder is in a double-bind, and there does not seem to be a way out of the design versus economics paradigm. If anything, independent establishments such as educational institutions and research groups have the scope to intervene and examine the situation with a broader perspective[22]. We are hopeful that such interventions will be put forward in the coming years, and through this project, we aim to move in this direction.

In another attempt to frame the problem, the authors[9] adopt medicalized, user-oriented, and design-oriented perspectives. From the medical perspective, some interactions with technology can be considered harmful and the behaviors can even be labeled as 'addictive', such as disordered use of internet gambling. However, such framing might lead to treatments in a medical setting where clinical standards are likely to exclude technology use altogether which in itself is problematic. From the user's perspective, the user's perceived loss of autonomy and control over the technology is the problem posed by digital devices.

This framing confines the issue to self-determination where workaround strategies are subjective in nature and cannot be considered as a global solution for the problem. Finally, the tech companies that employ the current interface in the first place have launched some additional features that allow users to set limits to their screen time. Such interventions are only superficial and reasonable so as businesses are being rewarded for holding users' attention. Tech companies even measure user engagement with metrics such as time spent on the product, the number of clicks, and scroll rate, and continually target to maximize them. As long as the companies measure their success with such metrics, their digital wellbeing interventions are bound to be superficial, existing without intending to support real wellbeing.

Technology abstinence via digital detoxes at first might seem to be the ultimate solution but, considering how valuable technology is in today's world of ubiquitous computing, Digital Wellbeing becomes arguably preferable to Digital Abstinence.

1.1.1 Design Paradigm - The Hook Model

To explain the addictive nature of digital devices, Nir Eyal, the author of *Hooked*[\[17\]](#), presents the concept of manufactured habits through hook cycles and refers to habits as automated behaviors triggered by situational cues, i.e., action with little or no conscious thought. The four-step hook model encompasses a trigger, an action, a variable reward, and an Investment.

Triggers are either external or internal. An external trigger is a call for immediate action from a user. For a digital gadget, external triggers include notifications, emails, messages, reminders, or even the static app icons on the home and menu screens. With subsequent encounters of external alerts, triggers are internalized to a routine. When a trigger is completely internalized, unprompted engagement is engineered.

The action step has to be as seamless as possible. The ease of performing the desired action became so crucial for the companies that it led to many innovations for interface usability and delight. Double-tap to like a picture, pull to refresh on social media timelines, long-press gesture for additional options, and infinite scrolls on pages are some usability innovations that have entered in the last few years and now are considered best practices in interface design with every application using them in their interface functionality. Hardly any conscious effort is demanded in performing an action through these gestures. Some actions are even automated like auto-playing the next recommended video on video-streaming services. Such automation lead to passive device usage. Motivation to act is

another component in this step. Even when usability is lacking, motivation can drive a user to accomplish an action.

Variable Rewards can best be compared to the gambling rewards of slot machines in a casino. While the expectation of a reward surges the levels of the neurotransmitter dopamine, adding a variability dimension multiplies it. Variable rewards are present in all habit-forming products in some form or the other.

The last step, Investment is about investing in the service or the product with an anticipation of reward in the long run. Investing time and effort to get acquainted with a product and become skilled, building a social profile with time and earning a reputation in the virtual social circle, submitting preferences and choices to services over time for obtaining better content, are all typical forms of investment in today's digital world. Users feed the companies with data or usage enabling the companies to get hold of them with triggers for the future. By increasingly coming back to the product, user's experiences become valuable and services become much harder to leave.

1.1.2 Persuasive Technology - Dark Patterns

Design in itself can be considered as an act of persuasion[19], and persuasive technology is capable of producing outcomes that harm the individual/society. A term invented by a UX practitioner Harry Brignull, Dark Patterns are tricks used in websites and apps that make one do things they didn't mean to, like buying or signing up for something. These are the instances where designers use their knowledge of human behavior to implement deceptive functionality that is not in the user's best interest[7][19]. Traditional dark patterns comprise of the tricks such as Bait and Switch, Disguised Ad, Forced Continuity, Trick Questions, to name a few. We would like to expand the concept to pervasive interaction controls that promote prolonged usage of digital platforms such as Infinite Scroll, Notification Dots, Pull to refresh, Autoplay Videos.

1.2 Digital Wellbeing

We are living our lives increasingly among digital devices such as smartphones, tablets, and laptops, which will go up further in the future with the invention of new devices such as wearables and VR sets. The businesses want to keep up with the competition by targeting continued attention and updating device technology accordingly. However, our



FIGURE 1.1: Extending Dark Patterns to Interactions

bodies and minds are limited and are not going to upgrade at the same pace. This can lead humans to lose control and become slaves of businesses disguised in technology.

As digital technologies permeate our everyday lives, it has become increasingly vital that we consider what impact this has on our wellbeing. Screen-based activities include work, study, socializing, leisure, entertainment, food gathering, traveling, sports, exercise, meditation, prayers, etc. On the upside, Digital technologies enable us to reap great benefits such as connectedness, pleasure, achievement, knowledge, and new opportunities. At the same time, they can lead us to huge risks such as avoidance, procrastination, stress, and even self-sabotaging. The capability of digital technologies to connect us together can paradoxically lead to disconnection in the physical world.

The time spent on digital devices in many instances is due to expectation, habit or boredom, and even FOMO (the fear of missing out). Our dependence is ever-increasing and so are the consequences. In order to take control and leverage the capabilities of technology, it is important to design technology as per the needs and limits of the human mind. This realization among industry's tech designers has led to a movement termed 'Digital Wellbeing'.

1.2.1 What is Wellbeing?

Wellbeing refers to a person's sense of self and the ability to live their lives as closely as the way they want to. Martin Seligman's PERMA model[42][43] suggests five basic elements of wellbeing: Positive Emotion, Engagement, Relationships, Meaning, and Achievement. According to the model, flourishing wellbeing is attained when one meets their potential,

has strong relationships with others, and does what's worthwhile to them. Wellbeing constitutes various kinds of wellbeing such as physical wellbeing, mental wellbeing, economic wellbeing, and emotional wellbeing, which are interlinked.

1.2.2 Definition(s) of Digital Wellbeing

As an experience: Digital wellbeing is a subjective individual experience of optimal balance between the benefits and drawbacks obtained from mobile connectivity.

This experiential state is comprised of affective and cognitive appraisals of the integration of digital connectivity into ordinary life. People achieve digital wellbeing when experiencing maximal controlled pleasure and functional support, together with minimal loss of control and functional impairment[46].

A narrow definition: The extent to which a person perceives their digital device use is aligned with their own long-term goals[23]. A broad definition of wellbeing includes domains of psychological wellbeing (life, satisfaction), education, community, health, work, environment, safety, etc.

1.2.3 Digital Evils

The saying from Socrates *"Thou shouldst eat to live; not live to eat."* can be applied to consumption of information as well. Information on digital devices is nothing less of a giant buffet open round the clock. A number of digital evils were identified in the Age of Information. Here are some evils relevant to the current study.

Fear of Missing Out (FoMO): The ability to remotely check email/news can help make more efficient use of time, but the fear of not seeing an email, or news story (or not seeing it promptly enough), compels increasing checking of digital devices. FoMO is characterized by the persistent desire to stay connected with digital settings with an apprehension that others might be having rewarding experiences from which one is absent[35]. FoMO can drive excessive checking of digital platforms, especially social networking sites, making it difficult to remain productive in daily life[16]. Stemming out from FoMO, compulsions of usage can be observed, such as

- Reaching out for digital devices without an actual need.
- The impulse to check a notification right away.

- Phantom Vibration and Phantom Ringing (Perception that a gadget is vibrating or ringing when it is not.)[14]
- The pull to visit a digital setting.
- While on the platform, repeatedly refreshing hoping to get new content.

Self Sabotaging: Subtle forms of self-sabotaging can be observed in behavioral compulsions of device usage.

- Passive scrolling on the newsfeeds and timelines for too long.
- A mindless routine of visiting a set of applications one after the other upon opening the digital device.
- Constantly comparing oneself to others (rather joys of others) on social media.

Aggressive forms of sabotaging can be observed with addictive syndromes such as Internet Addiction Disorder, Internet Gaming Disorder, Television Addiction /Streaming Addiction, where users are trapped in problematic usage.

Prolonged Use: Collectively captured under ‘*Computer Vision Syndrome*’, excessive screen time and intense use of gadgets lead to undesirable effects on multiple aspects of wellbeing such as sleep cycles, alertness during waking hours, fatigue, physical strain, and cognitive performance.[45]. Prolonged use of computer-mediated communication platforms has led to negative experiences, and ‘*Zoom fatigue*’ is the most prominent example. The exhaustion is mental as well as physical. The interaction and communication exchanges of a face-to-face setting exhaust people when performed in a virtual setting.[29]

Communication: Specific to messaging, users can experience a variety of imposed emotions. As a sender, users can experience anxiety while waiting for a response and even more when there is a delay in receiving a response. The anxiety arises from the assumption that the other party is available and that they are choosing not to respond/revert. As a receiver, there is an induced obligation to respond at the earliest.

Filter Bubble: Selective information segregation is a serious concern, and ‘Filter Bubbles’ assigned by personalized content propagate it by reinforcing one’s own views, beliefs, and opinions. Machine learning models inadvertently create bubbles while attempting to provide the content an individual will interact with.[18] Such bubbles can expand among individuals to a group of like-minded people to a neighborhood and eventually amplify the diversification of ideologies to the population at large.

1.3 Current Study

1.3.1 A Design Choice - Smartphone Menu

A number of design choices make up today’s ubiquitous UI on smartphones. We identify the digital wellbeing concerns of some design choices such as Infinite Scroll, Navigation Dots, and Pull to Refresh. Such interactions can be discarded from use as they are not mandatory for functionality. Our main focus will be on a more fundamental design choice - the smartphone menu grid of app icons. The alternative to the menu grid is a search field.

The design principle that backs up the pervasive use of menu grid is a usability heuristic ‘Recognition rather than Recall’[31]. The 10 principles of Heuristic evaluation are developed as a “discount usability engineering” method for evaluating UI on digital devices. The principle explicitly covers the items - See-and-point instead of remember-and-type, provide list of choices, show icons and other visual indicators, among others[30].

Through our study, we challenge this principle under a set context of app selection interaction on smartphones.

1.3.2 A Digital Wellbeing Issue - Goal Slippage

Digital Wellbeing is concerned with the alignment of digital device usage and users’ goals. It encompasses a range of issues. We specifically address a Working Memory issue called Goal Slippage. Goal Slippage is the failure to return to the primary task after attending to an interruption task. We think that the aforementioned design principle has created an Interface design that is detrimental to the Digital Wellbeing of the end user.

1.3.3 Conceptual Basis

Miyata and Norman[28] suggested two kinds of conflicting properties exhibited by the human information processing system, namely continual concentration and continual distraction, which lead to task-driven processing and interrupt-driven processing, respectively. In a task-driven state, people get deeply involved with their primary tasks and become insensitive to external interruptions. In an interrupt-driven state, people seek extraneous events and are easily distracted by interruptions as they do not have a structured task/-goal. Miyata and Norman claim that it is difficult to maintain a task-driven state in the

presence of external events irrelevant to the primary task. For task-driven processing to continue, the schemas relevant to the primary task should dominate. Based on this conceptual segregation of information processing, we believe that the device interface design should cater to different kinds of processing differently.

1.3.4 Problem Statement

A digital gadget user should be able to use their gadget for their needs and solely for it. In today's world, everyone uses their device for more than just that. We daily spend 3 hours and 15 minutes on our phones[25], a number that can easily shoot up considering the post-pandemic times. We experience a lack of control over our usage[24]. A part of the problem is deliberate design. Interfaces are primarily interrupt-driven by design. Can we pursue a task-driven design approach and undo the disruptive nature of gadgets? We think that the current best practices and principles of Interface design will not lead there.

To establish empirical evidence, we set out to capture the extent of Goal Slippage and Loss of Content with the altering presence of external interruptions and distractions. For the level of distractions, our study challenges the pervasive menu-grid interaction for smartphone app selection and thereby challenges the heuristic design principle 'Recognition rather than Recall' that recommends it. We present plain search as an alternative. For the level of interruptions, a mandatory interruption task is employed to represent the interruption driven usage against task driven usage. If results support our line of prediction, we will propose a set of design principles and attempt to design an interface for a smartphone launcher that mitigates goal slippage.

1.3.5 Outline

In Chapter One, the background for the study has been described, and the concept of Digital Wellbeing was introduced and the research objectives have been identified.

In Chapter Two, goal slippage is introduced and the existing literature on interruptions is reviewed.

In Chapter Three, the hypotheses are framed and details of the experiment that was conducted are presented. The chosen experimental approach is also justified.

In Chapter Four, the data from the experiment is analysed. The results point us towards an alternative design.

In Chapter Five, a Design System that re-imagines the smartphone pick up(or launch) experience is proposed guided by the design process and the results in chapter four.

Chapter 2

Interruptions, Distractions, and Goal Slippage

In order to achieve Digital Wellbeing by Design, it is crucial to set up the distinction between the status quo and the target state. In digital devices such as smartphones, Digital Wellbeing endeavors to provide complete agency to the user serving their own interest. Current User Interfaces(UI) demand attention for many elements on the interface without considering the user's goal. Thus, digital device usage is often reactive in nature, i.e., user interaction is driven by the device's suggestions rather than the user herself. For instance, attention seeking UI at the system level in the form of notifications that welcome the user when they turn on the device, UI at an application level in the form of a newsfeed on a social media platform when they sign in.

2.1 Distraction

According to the interruptions literature, anything demanding attention from the user that is not associated with the user's primary goal at a given moment is a Distraction. Distractions involve momentary lapses of attention on the primary task without an active engagement in the interruption task[11]. Clapp et.al [10] defined distractions as entirely irrelevant stimuli that should to be ignored.

According to Oxford dictionary, the word 'Distract' means prevent someone from concentrating on something and in comparison, the word 'Interrupt' means stopping/breaking the continuous progress of an activity or process.

2.2 Interruptions

An Interruption is defined as the suspension of one stream of work prior to completion, with the intent of returning to and completing the original stream of work[3]. In similar line, Clapp et.al [10] defined interruptions as interfering stimuli that necessitate attention such as a secondary task . For instance, attending to phone calls while pursuing a primary task can be an example of an interruption as it needs to be performed whereas something like a background noise while performing a primary task can be a distraction.

Task switching and distractions, in general, are associated with increased error rates. As per the taxonomy of error, the failure to return to the primary task after attending to an interruption task is classified as a goal slip(in Medical Errors)[47] or a loss of activation slip(in HCI)[33] which we refer to as ‘**goal slippage**.’

Human Working Memory is limited in capacity; that is, it can only process a certain amount of information at once[27][12]. A stance in multitasking goes so far as to claim that simultaneous performance of non-automatic multiple tasks is only a perception arising out of repeated task switching[44]. When interruptions are at play, they can completely hijack the working memory above the interference level. They can result in “What am I doing here?” experiences[38] which translate to “Why did I turn to my device?” at a system level interface or “What did I open this application for?” at an application level interface, if at all there is a realization left that there was an actual goal that was not executed.

Interruptions can be triggered either by external nudges or internally. Previous research shows that humans internalize their patterns of interruptions and trigger them without the need of a nudge or an environmental cue[13][17]. This is not a surprise because of the implementation of habit-forming systems. It is not by accident but by deliberate design. If one still aims to achieve the broader goal of Digital Wellbeing, one has to do away with the current best practices in UI Design and evaluate any Interface Design element under the lenses of Usability and Agency.

Before proceeding with the Design of the system, we want to establish empirical evidence for the current state of goal slippage - whether it exists and to what extent it occurs. Only then can we advocate a design that reduces or preferably eliminates goal slippage. To our knowledge, no such evidence exists in the HCI literature or in the interruptions literature of applied cognitive science.

2.3 Findings from Interruptions literature

Most studies insist that Interrupting a task decreases the general performance barring a few studies reporting benefits on the task itself, such as effects on creativity[11]. Time on Task and Resumption Lag variables were the most represented variables to measure the effects of interruptions while comparing interrupted to uninterrupted trials. A goal is a representation of a person's intention to accomplish a task, and the necessary resources for different processes are provided by attention, which is limited[21], suggesting that it be invested cautiously. While shifting to an interruption task from the primary task, it is suggested that the components involved in the primary task or the associated schemata completely extinguish in 30-seconds when active maintenance is not allowed.[11]. The complexity of the primary task or the interruption task is supposed to have an effect, but the direction of the effect is not consistently established theoretically or with empirical research. Interruption effects could also be dependent on the timing of the interruption.

When interrupted, an average knowledge worker nearly takes 30 minutes to resume the original task[1]. Some IT professionals interrupt themselves in just 20 seconds of attending a primary task[8]. Interruption frequencies are associated with stress levels. Cutting off from constant notifications such as emails were found to have significant differences in stress levels[8]. However, some studies point out that self-interruptions to check emails increase in the absence of external notifications[20].

2.4 Goal Architecture

Theoretically, with some debate, goal storage is mapped to the Stack data structure of computation[2]. The storage of Goals of complex hierarchical tasks were explained with the Stack's 'last in, first out' behavior. This allowed explanation of complex problem solving with limited working memory capacity by using means-ends analysis. A complex task is decomposed into a set of multilayered subgoals until they reach a tractable size that can be achieved directly. The same architecture is built into the ACT-R framework for goal storage processes.

The Memory for Goals model proposed by Altmann and Trafton[2] describes the process of encoding and resuming the goal after an interruption by employing activation and associative priming as stated below. At any point in time, the most activated goal(primary goal) drives behavior. In order to become the primary goal, the activation level of the goal

has to be above the interference level. The interference level is equivalent to the mental clutter formed by distractions, old residual goals. A new goal quickly reaches its peak activation level due to higher initial sampling. However, the peak can not be sustained due to the system's involvement in the action. The activation level deteriorates as a function of Time, and when deliberate interventions of activation are absent, it falls below the interference level, and consequently, a new goal is activated. This decay demands a periodic re-evaluation of the system's cognitive direction. A suspended goal is masked by residual interference and the new primary goal. At this moment, only knowledge or environmental cues can re-activate the suspended old goal. For this activation, the cue has to be obvious to decode, and the association between the cue and the respective goal has to be made.

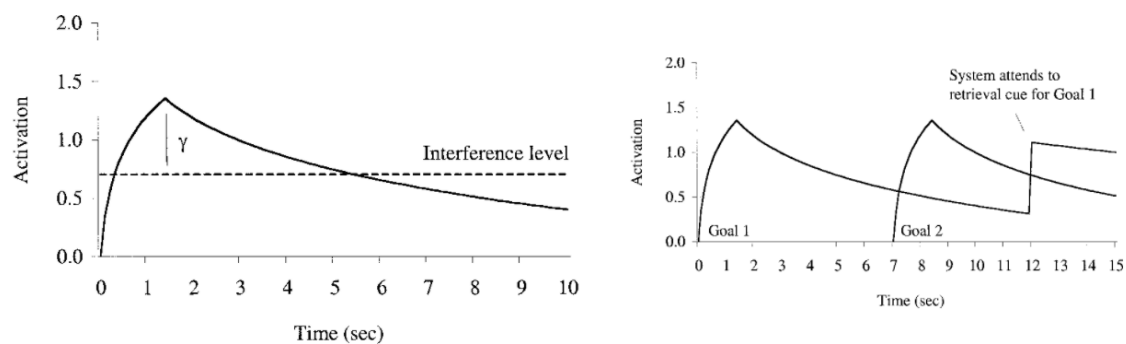


FIGURE 2.1: Memory for Goals model

Let's apply this architecture to the launch experience of smartphones. The user's primary goal has led them to reach out to their device. An external interruption in the form of a notification is activated and deteriorates the primary goal as time progresses. The original goal is masked at the interference level. Since the relevant action is not executed on the device, no trace of it in the form of the cue is present in the interface. Thus, goal slippage is likely to happen in the absence of knowledge cues.

2.4.1 Prospective Memory

Prospective Memory is defined either as remembering to perform an action at a particular moment in the future or as retrieval of planned intentions and their execution. Failures in Prospective Memory imply failures in the retrieval of an intended action at an appropriate moment[5]. There are four stages of prospective memory for successful implementation. They are Encoding, Storage, Retrieval, and Execution.

2.4.1.1 Attributes of a delayed intention

As the definition of Prospective Memory revolves around the concept of ‘Intention,’ it is important to classify its types and attributes. Depending on execution, an intention can be either immediate or delayed. Immediate intentions allow the execution of intentions right away, whereas delayed intentions demand an interval before their execution. Behaviors involving delayed intentions fall under Prospective Memory. Variations of ‘to be remembered’ intentions are theoretically classified at each stage of Prospective Memory[5]. A sample classification of attributes of an intention is given below.

Intention: Telephoning a close friend in the evening, after watching a TV Program to know his whereabouts

- Simple Decision/Thought (Assuming that there are no competing tasks)
- Self-generated
- Unimportant (A casual conversation)
- Pleasant
- Time-based (4 PM)
- Activity-based (After watching a TV program)
- External Cue is absent (Relying solely on knowledge-based cue)
- No interruption at the time of retrieval. (Right after an activity is finished)
- Time & Activity Combined
- Episodic & Infrequent
- An intermediate window of opportunity (5-10 mins gap is allowed)
- Short Term Execution (Expected Call Duration is 15 minutes. A few ‘to be conversed’ topics might be planned and encoded)
- One Stage Execution (Call the person directly from the contact list)

2.5 Goal Slippage in Digital Devices

Goal slippage can occur with tasks of Prospective Memory or due to pure Task Switching. The difference between the two is execution. If an intention solely exists in the user's mind without any external presence in the environment, it becomes a task of Prospective Memory waiting to be executed at an appropriate moment in the future. If the intention is executed and interruption disrupts it, it is pure Task Switching, whose cognitive processing can be best explained by the Interruption Research Literature.

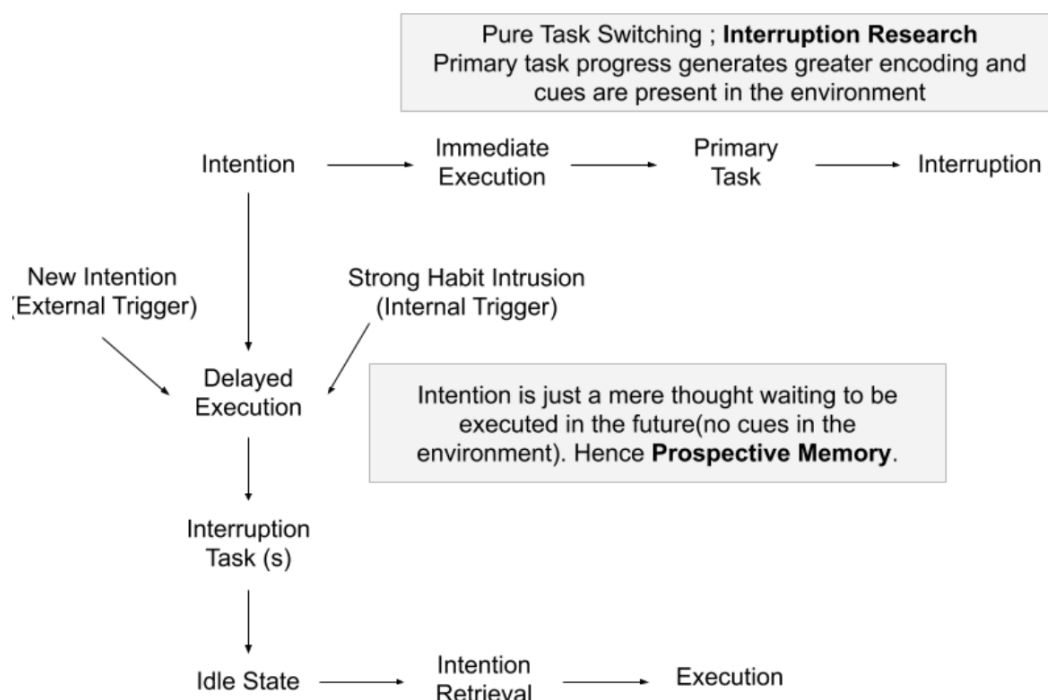


FIGURE 2.2: Immediate and Delayed Intention

Interrupted intentions or goals during the smartphone launch will convert the immediate execution of an intention to a Prospective Memory function. Encoding could be strengthened at the moment of interruption. If not, the user will have to rely on the past encoding for the particular intention. As the user turns to the interruption task(s), the second stage, 'storage', is initiated, wherein the user has to store the intention till an appropriate moment for retrieval. After the completion of the interruption, the user enters the retrieval stage and relies on cues either in the environment or in their own knowledge. For a cue to be effective, it has to be decoded into the intention. Once the intention is available to the user, the action associated with the intention is to be followed for execution.

We propose that task-based processing should be supported by the Interface design. When a user turns towards a digital device with their own intention or goal, it has to be strengthened by encoding before the user attends to any interruptions. The interruptions could be external (in the form of notifications) or internal (strong habit intrusions). If these interruptions are to drive the behavior or user interaction, it falls under interrupt-driven processing. Interruptions are not necessarily bad or useless, and thus, we cannot eliminate them completely. A balance between them would be essential for Interface design. We propose that this balance can be achieved by revising the order of user intention and interruptions. While a user picks up the smartphone and turns it on, the smartphone UI can be designed to let user note-down their task somewhere on the home screen or better if the interface promotes the user to execute their task right away. Thus, even if interruptions are to be acknowledged or attended thereafter, the user has already successfully encoded or initiated the execution of their intention. This activity (even if it is incomplete) will leave enough external cues that users can later rely on to recall the original intention and continue to execute the associated task.

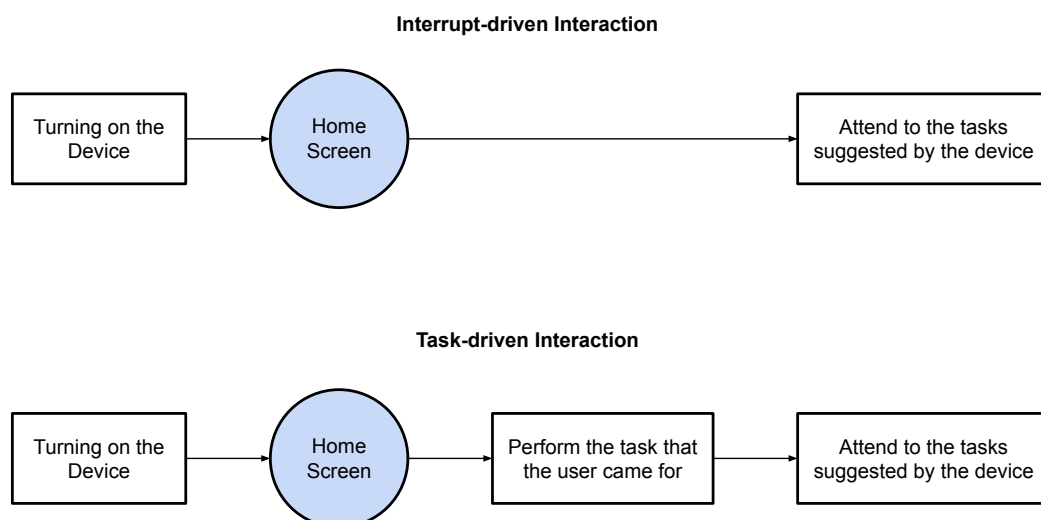


FIGURE 2.3: Types of Processing supported by Smartphone Launcher

Chapter 3

Experiment

This chapter details the testable hypothesis, experiment design, method, and measures captured to analyze the results.

3.1 Hypothesis

While interacting with a smartphone for accomplishing a specific goal/task, loss of content and goal slippage is Significant when *the primary task is externally interrupted*.

While interacting with a smartphone for accomplishing a specific goal/task, loss of content and goal slippage is Significant when *distractions are present*.

3.2 Experiment Design

To investigate the phenomenon of Goal Slippage in digital devices, we designed an experiment that captures the extent of Goal Slippage with the altering presence of Interruptions and Distractions. The experiment mimics the standard smartphone launch experience of the real-world setting, where the user reaches for their device to accomplish a specific task. Typical device distractions while opening a device can be considered to be anything on the device that is not in line with the user's actual task at hand. It can be the notifications that appear upon opening the device, alerts on the notification bar, the icons on the screen, and even the background picture. Any of these distractions could become an interruption once the user engages with it. Since the notifications that appear on opening

the device typically are of significance to the user, the user can get engaged with some of them while the original task is held in memory, waiting to be executed after completing the interruption task. Most used application icons present on the screen could also be important because of habitual usage and can engage users and lead to interruptions. However, which distraction is of significance to the user is subjective and varied. Also, in the real world setting, no one is watching the user and in some sense, the choice of which task to perform is with the user i.e., whether to engage or not with the interruption task. Interruptions can be classified into two categories, internal and external. While internal interruptions are triggered by the user, external interruptions arise from the user's environment and are located outside the user. Due to the limitations of the experiment setting, we employ mandatory interruption tasks in our experiment design which are external interruptions. We acknowledge that the nature of interruptions in our experiment are not exact representations of those present in the real-world.

Each trial of the experiment is initiated with a need to reach out to the device in the form of the primary task and this primary task is setup for a recall paradigm. Unlike a typical interruptions paradigm where participants engage with the primary task, participants in our experiment were only asked to memorize the primary tasks. The following choices were made in the experiment design.

- All the primary tasks (see Appendix A) were handpicked to closely resemble the tasks that users perform with their devices on a day-to-day basis. Each task specified a target application for executing it and had two items of task content, as illustrated below. To successfully recall the task, a participant had to recall all three keywords.

Book a **cab** to the **railway station** on **Uber** App

Keyword 1 Keyword 2 Keyword 3

FIGURE 3.1: Framing of the primary task

- A time-bound modified version of an anagram-making word game was chosen as the interruption (external) task. For the external interruption absent variant, a blank screen with a fixation cross was presented for the same duration as the external interruption present variant. Thus, time was controlled. However minimal in presence, a blank screen doesn't constrain mind wandering (Internal Interruption). For this

stage, displaying any kind of timer is deliberately prohibited so as to control for any anticipation effects.

- The distraction-present variant had a menu grid interface with a 7 x 5 grid where every app icon other than the target icon acted as a distraction. The menu grid height was adaptable in order to fit within the device screen height so that scrolling was eliminated. All app icons were displayed along with the app names in alphabetical order. The distraction-absent variant had a standard text based search interface.

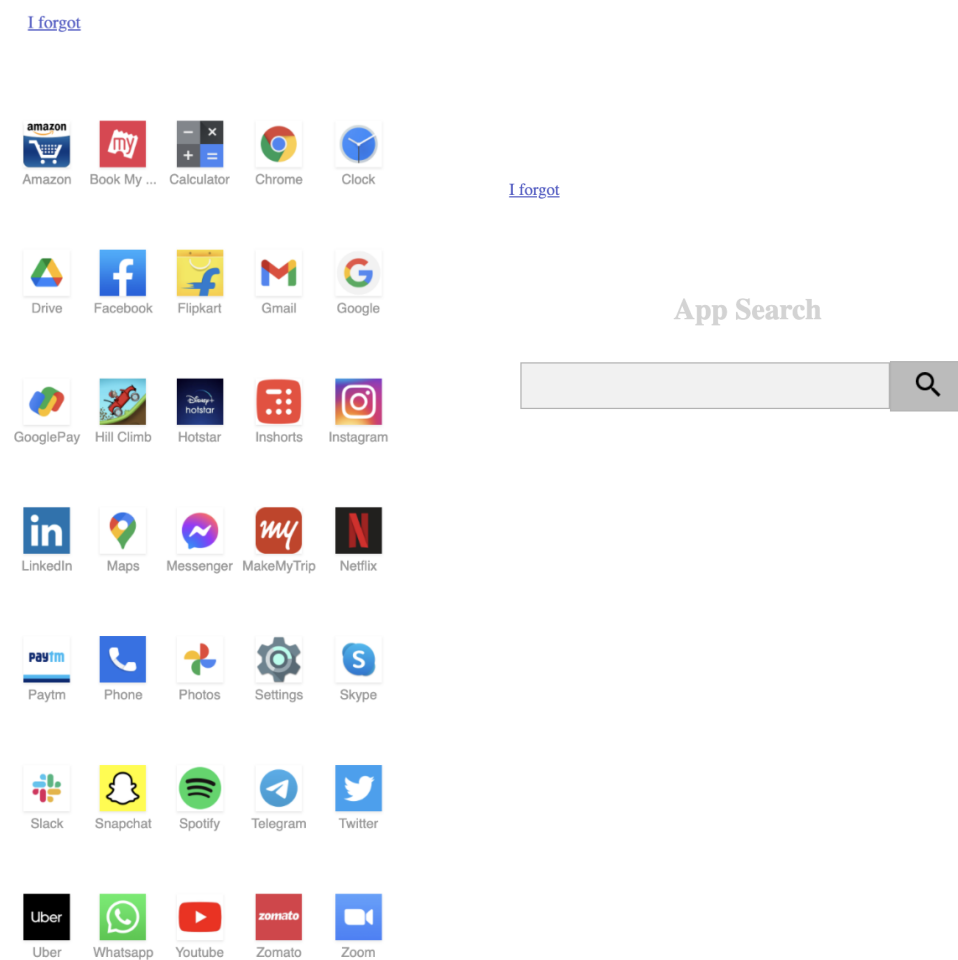


FIGURE 3.2: Distraction-present(left) and Distraction-absent(right) variants

3.2.1 Experiment Links

Instructions and Practice Trials: <https://www.cgs.iitk.ac.in/wwwrw/sruzan/Exp/>

Main Experiment: <https://www.cgs.iitk.ac.in/wwwrw/sruzan/Exp/task.html>

Survey on Task Variability: <https://forms.gle/W9ZmPeTT7yi2pA2t6>

3.3 Method

3.3.1 Participants

Forty-two participants (17 female; mean age = 27.14, age range = 21–38) were randomly selected and recruited for the study. Participants were either graduate students or working professionals from varied backgrounds. To qualify for the study, participants had to possess a smartphone gadget and be fluent in English. They received a gift voucher worth ₹100 for their participation.

3.3.2 Pilot Study

A within-subjects design was employed where two factors acted as independent variables with two levels each. The factors were ‘Interruption’ and ‘Distraction’ with levels being presence and absence. We ran two pilot runs with four and ten participants, respectively. With the first pilot run, we ensured that participants understand the task paradigm. During this run, the interruption task duration (10 seconds) was found to be short for capturing goal slippage. In the second pilot run, we tested with a duration of 18 seconds and found a significant effect size for interruption. The same was used for the main experiment. We determined our sample size for main experiment from power analysis of ANOVA main effects and interaction (0.1 Cohen’s f , 0.05 α , 0.95 β) for 2×2 factorial within-subjects design.

3.3.3 Ethical Review

The Institute Ethics Committee(IEC) of Indian Institute of Technology Kanpur approved the study.

3.3.4 Procedure

The experiment was conducted online and consisted of one 45-minute long session. Participants were asked to join a teleconference call and were provided instructions (see Appendix B). They were familiarized with the task paradigm through two practice trials. The practice session was monitored.

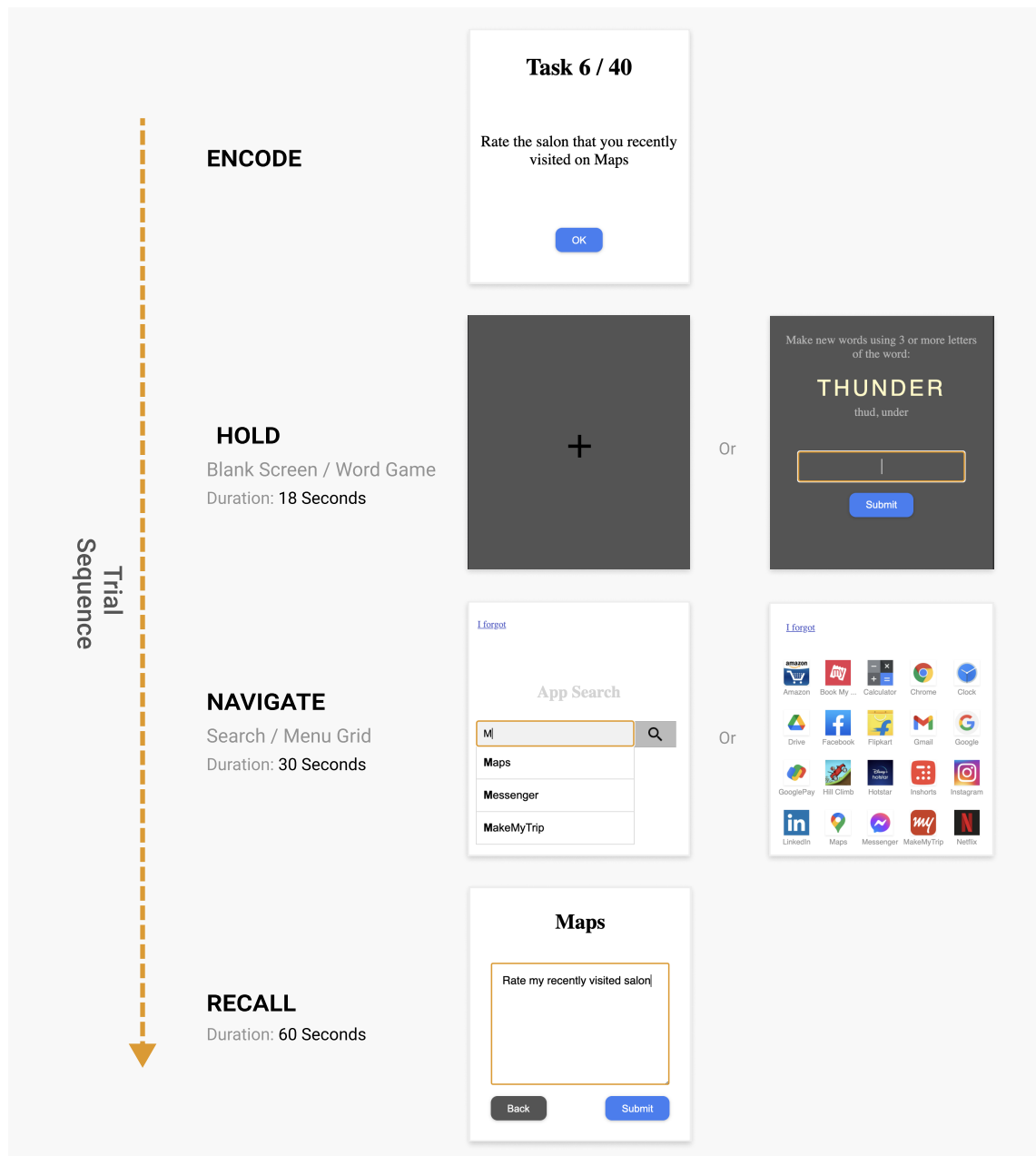


FIGURE 3.3: Trial Sequence(Stages from Top to Bottom)

Each trial consisted of four stages, Encoding, Hold, Navigate, and Recall. During the ‘Encoding’ stage, participants were given a task and asked to memorize the details of the task. They were encouraged to take as long as they needed to memorize the task. For the first three seconds, the next step CTA button was disabled to eliminate unintentional clicking. The second stage, ‘Hold’, had two randomised possibilities - either a Blank Screen or a Word Game. The blank Screen had a fixation cross appearing in the center, and participants were instructed not to look away from the screen. And when the word game appeared, participants were instructed to do their best and submit as many words as they could. The word game was a sub-anagram word-making game where new words are formed with three or more letters from a given word. For example, the word ‘THUNDER’ can be rearranged to form words such as THE, DEN, TEN, HEN, RENT, DENT, UNDER, etc. Each word appearing in the word game had at least twenty possible sub-words. After eighteen seconds, the trial automatically proceeded to the ‘Navigate’ stage, where participants had to recall the target app for accomplishing their task given in the ‘Encoding’ stage. Either of the two interfaces - Search or Menu Grid would appear in this stage. For the search component, suggestions were provided upon typing. In the Menu Grid, all app icons (36×36 dimension) were displayed along with the app names in alphabetical order. A button ‘I forgot’ was provided in this stage, and participants were instructed to click it in case they forgot the task. There was a thirty-second time limit for this stage beyond which, the trial ended and moved to the next one. If the participant selected an app within the time limit, they moved to the last stage, ‘Recall’. In this stage, participants had to recall the task and manually input it in the text box provided. The time limit for this stage was sixty seconds. Participants were told that they need not reproduce the given task with as-is phrasing but to report the contents of the task. The experiment consisted of forty trials, and participants were instructed not to take any breaks in between. The entire experiment session interaction was recorded using a screen recording tool.

After completing the trials, participants rated each task based on their familiarity and pleasantness. Both the ratings were taken on a 5-point Likert scale where 1 represented least and 5 represented most. Participants were then asked to provide feedback. They were asked to report any trials where they found the tasks to recall to be hard and state the reasons from their experience.

3.4 Measures

3.4.1 Demographics

Along with age and gender, participants were asked to report their average estimated time spent on their smartphones in a day. They were provided with four choices: i) less than 3 hours, ii) 3-4 hours, iii) 4-5 hours, and iv) more than 5 hours. We call this data variable *Screen Time*. Participants were also asked to report the approximate number of apps installed on their smartphone. The four choices were: i) less than 40, ii) 40-80, iii) 80-120, iv) more than 120 apps. The start time of participation was recorded.

3.4.2 Explanatory Variables and Levels

A within-subjects design was employed where two factors acted as independent variables with two levels each. The factors were ‘Interruption’ and ‘Distraction’ with levels being presence and absence. In the ‘Hold’ stage a Blank Screen meant the absence of Interruption and the Word Game indicated the presence of Interruption. The Search component during the ‘Navigate’ stage meant the absence of Distraction, and the Menu Grid meant the presence of Distraction. For each participant, 25% of the trials belonged to each condition of the 2×2 factorial design.

Factor	Levels	
	Absence of factor	Presence of factor
Interruption	Blank Screen	Word Game
Distraction	Search	Menu Grid

TABLE 3.1: Factors and their Levels

Data points from each participant = 40

Number of Participants = 42

Total Data Points = $42 \times 40 = 1680$

Data Points in each condition = $1680/4 = 420$

The four conditions are (Blank Screen, Search), (Blank Screen, Menu Grid), (Word Game, Search) and (Word Game, Menu Grid).

3.4.3 Dependent Variable - RT Goal

RT Goal is the response time in the ‘Navigate’ stage, i.e., it is the time taken to reach the target application via the search or menu grid UI component. The target application is the one that is specified in the ‘Encoding’ stage of that trial. Resumption Lag as a dependent variable is suggested to represent the ongoing effects of interruption better[11]. The measure RT Goal is the closest measure to resumption lag as we can not distinguish precisely when the search action is being carried out at Navigate Stage.

3.4.4 Task’s Subjective Variability

Familiarity and Pleasantness ratings were captured for each task which represents the inherent variability of a task to a participant. Familiarity captures how familiar, i.e., how often the participant performs the given task on a 5-point Likert scale. Pleasantness captures how pleasant the given task is to the participant, i.e., whether the task itself is boring or delightful, on a 5-point Likert scale.

3.4.5 Other Variables

RT Encode, the time taken to memorize the task during the ‘Encode’ stage, and RT Content, the time taken to manually enter the content, were captured for each trial.

Chapter 4

Results and Discussion

This chapter presents all the results of analysing the data from our experiment and discusses our interpretation of the results. We first report the instances of *Goal Slippage* and *Loss of Content* per condition, and analyse the differences, followed by RT Goal Distributions of each condition. We then report our main result i.e., Two Way ANOVA of RT Goal as a function of Interruption and Distraction Factors, analyse the differences in variances and right-tails, followed by effects of Task attributes, effect of RT Encode on RT Goal, and end the section with an exploration of Demographic Effects.

4.1 Results and Analysis

4.1.1 Loss of Content and Goal Slippage in each condition

The trials where participants could not recall the goal (i.e. the target app) are assigned under ‘Goal Slippage’. The trials where the goal is recalled but content keywords were either not recalled or recalled incorrectly are assigned under ‘Goal Retention but Loss of Content’. Participants’ recall is presented below out of the 420 trials in each condition.

	Blank Screen		Word Game	
	Search	Menu	Search	Menu
Goal Slippage	1	3	19	18
Goal Retention but Loss of Content	19	25	27	36
Complete Retention	400	392	374	366

TABLE 4.1: Condition-wise Goal Slippage and Loss of Content

The difference in the number of trials under each category of Goal Retention (Goal Slippage, Loss of Content, and Complete Retention) to each condition is tested with chi-square tests. The number of trials assigned differed significantly in the interruption conditions (Blank Screen, Word Game), but did not differ in the distraction conditions (Search, Menu).

	Blank Screen			
	Search	Menu	chi-square	p-value
Goal Slippage	1	3	1.89	0.39
Loss of Content	19	25		
Complete Retention	400	392		

	Word Game			
	Search	Menu	chi-square	p-value
Goal Slippage	19	18	1.39	0.49
Loss of Content	27	36		
Complete Retention	374	366		

	Search			
	Blank Screen	Word Game	chi-square	p-value
Goal Slippage	1	19	18.46	0.0***
Loss of Content	19	27		
Complete Retention	400	374		

	Menu			
	Blank Screen	Word Game	chi-square	p-value
Goal Slippage	3	18	13.58	0.001**
Loss of Content	25	36		
Complete Retention	392	366		

TABLE 4.2: Chi-square tests between interruption and distraction conditions. (Bold indicates significance at Bonferroni alpha of 0.00625)

While we did not test for this condition we assume that if a participant was asked to immediately execute a task after the encoding period he/she will neither exhibit goal slippage nor content loss. To test for the significance of the difference in the number of assigned trials with immediate recall, we have assumed the following baseline assignment: Goal Slippage: 0, Loss of Content: 0, Complete Retention: 420. This is, of course, a post hoc analysis and done under the above assumption. Under no disruption and immediate recall, it is reasonable to assume that there is no Goal Slippage or Loss of Content. Note that, in our experiment, we have let the subjects take their own time for encoding a task.

With this baseline as the reference, the number of assigned trials significantly differ in each condition, with the closest one being Blank Screen and Search Category with $\chi^2 =$

20.48, $p < 0.001$. Bonferroni alpha for all the comparison pairs(8) is adjusted to 0.00625. The test results are shown below.

	Baseline	Blank Screen & Search	chi-square	p-value
Goal Slippage	0	1	20.48	0.0***
Loss of Content	0	19		
Complete Retention	420	400		

	Baseline	Blank Screen & Menu	chi-square	p-value
Goal Slippage	0	3	28.96	0.0***
Loss of Content	0	25		
Complete Retention	420	392		

	Baseline	Word Game & Search	chi-square	p-value
Goal Slippage	0	19	48.66	0.0***
Loss of Content	0	27		
Complete Retention	420	374		

	Baseline	Word Game & Menu	chi-square	p-value
Goal Slippage	0	18	57.71	0.0***
Loss of Content	0	36		
Complete Retention	420	366		

TABLE 4.3: Chi-square test for each condition with the assumed baseline. (Bold indicates significance at Bonferroni alpha of 0.00625)

4.1.2 Distributions

The Probability Distribution (PDF) and Cumulative Distribution Functions (CDF) for each condition encompassing 420 data points are plotted to compare individual RT Goal distributions.

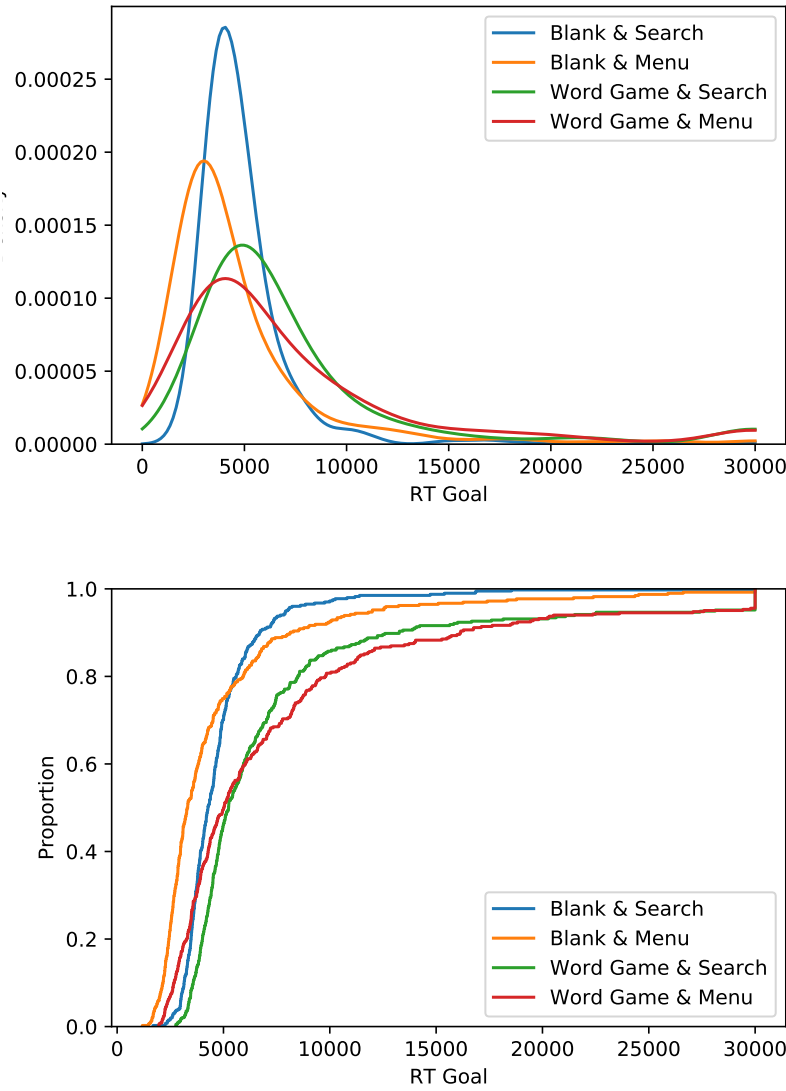


FIGURE 4.1: Probability Density and Cumulative Density Functions of condition-wise RT GOAL

Skewness and Kurtosis of the overall RT Goal distribution are 2.9 and 9.3 respectively and that violates normality. This implies that we should not be using F-tests, T-tests and ANOVA. However, we rely on empirical findings that tested the robustness of statistical tests when the normality assumption is violated[40][4] and the fact that sample sizes for

the data sets are fairly large and sphericity holds (as there is only 1 degree of freedom) and do our analysis, anyway. As outliers are important for our study and as there is no valid reason to observe random outliers only at a specific condition, we do not make any corrections or transformations to discard them or de-emphasize them.

4.1.3 Two Way Repeated Measures ANOVA of Average RT Goal

RT Goal was analysed using a 2×2 repeated measures ANOVA with the within-subject factors *Interruption* (Blank Screen, Word Game) and *Distraction* (Search, Menu Grid) to evaluate whether they modulated RT Goal at a significant level.

For trials assigned under ‘Goal Slippage,’ an RT Goal of 30 seconds was assigned, and they were included in all the analyses. The trials under ‘Goal Retention but Loss of Content’ were excluded.

The ANOVA was carried for RT Goal averaged across all the trials per each subject. Each subject had a corresponding RT Goal value in each condition. 42 observations were present in each condition.

Source	SS	DF	MS	F	p-value	η_p^2	Cohen's f (Effect Size)
Interruption	3.4e+08	1	3.4e+08	74.3	0.0***	0.3118	0.67
Distraction	4.1e+03	1	4.1e+03	0.0	0.98	0.0	0.0
INT * DIS	2e+05	1	2e+05	0.04	0.83	0.0	0.0
Residual	7.5e+08	164	4.6e+06				

TABLE 4.4: Two Way ANOVA - Interruption and Distraction on Average RT Goal per subject - Excluding the trials under Loss of Content

Large Effect (0.67 Cohen's f) of *Interruption* was observed, $F(1,164) = 74.3$, $p < 0.001$. *Distraction* and *Interactions* did not have any significant effect on RT Goal.

4.1.4 (Post hoc) Variances between Distraction Conditions

From the complete distributions of RT Goal (figure 4.1), the differences between the distributions of the two distraction conditions are evident. The main ANOVA could not test for any differences in the distributions beyond the mean. For instance, we observe that, ‘Blank & Search’ has a taller peak and a limited spread with hardly any RT Goal values present after 12 seconds whereas the ‘Blank & Menu’ condition has an early peak with a

long tail. Early peak and the long tail are averaging out when only the mean data is considered. In order to test for further differences we performed post-hoc analysis on variance differences between the two distraction conditions. Condition-wise standard deviations and variances are reported below.

	Blank Screen (INT =0)		Word Game (INT =1)	
	Search (DIS =0)	Menu (DIS =1)	Search (DIS =0)	Menu (DIS =1)
Standard Deviation of RT Goal	2.55	4.32	6.20	6.61
Variance of RT Goal (in seconds ²)	6.52	18.68	38.39	43.73

To test for the equality of variances of RT Goal between the two distraction conditions an F test was conducted. Under Blank Screen condition (no external interruption), variances were found to be significantly different between distraction conditions with F value 2.865 ($p < 0.001$, $df_1 = 419$, $df_2 = 419$). In the word game condition variances were not significantly different from each other with F value 1.139 ($p = 0.09$, $df_1 = 419$, $df_2 = 419$).

4.1.5 (Post hoc) Subject-wise variances between Distraction Conditions

The previous comparison assumes that each RT Goal is coming from a different subject. To test for equality of subject-wise variances between distraction conditions, we captured the variances of RT Goal for every subject across all conditions. The variance distributions made up of 42 values for each condition did not pass the Shapiro-Wilks Normality test. Thus, a non-parametric test, paired Wilcoxin test was performed to evaluate the difference in variances between the distraction conditions.

Under the Blank Screen condition, subject-wise variance distributions differed significantly between the two distraction conditions with p-value < 0.001 ($w = 105$). But, under word game condition, RT Goal distributions did not differ significantly ($w = 382$, p-value = 0.38).

4.1.6 (Post hoc) Subject-wise Right Extremes between Distraction Conditions

Since the focus of interest is the right tails of the distributions, we compared the largest RT Goal values between the two distraction conditions for each subject. Forty two values are considered for each condition.

Complete Retention	Blank Screen		Word Game	
	Search	Menu	Search	Menu
Mean of largest RT Goal (in seconds)	7.6	11.7	12.7	14.8
Variance of largest RT Goal (in seconds ²)	9.34	41.3	31.9	46.0

The distributions of these right-extreme RT Goal values are plotted below. The difference in means between the distraction conditions becomes evident from these plots.

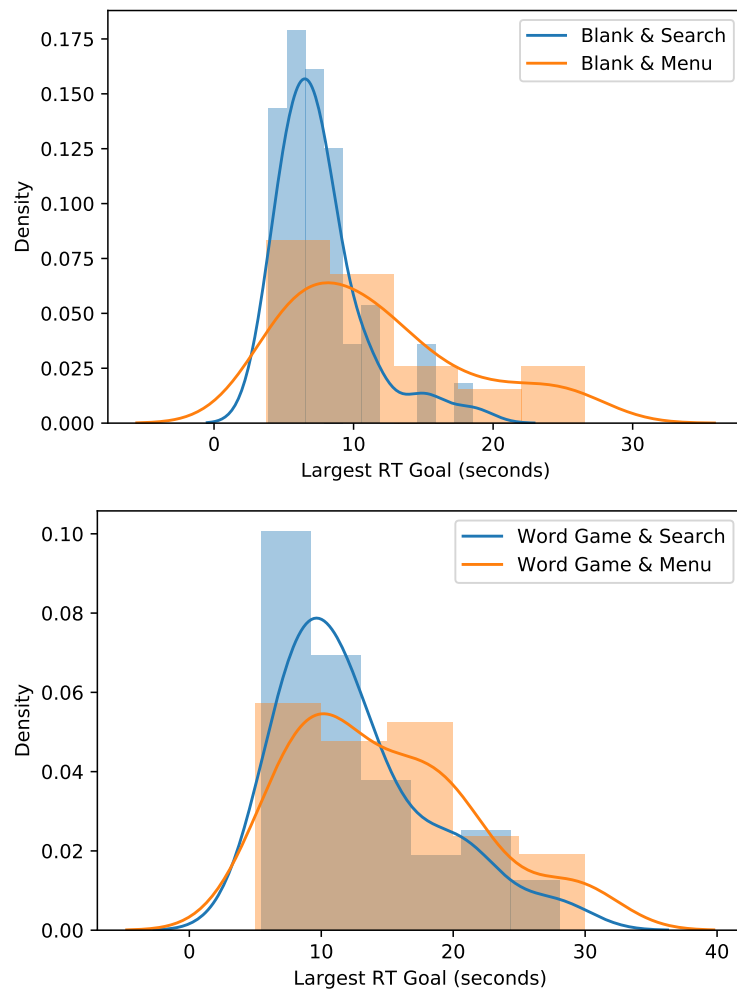


FIGURE 4.2: Right-end tails

None of the four distributions passed the Shapiro-Wilks Normality test and sample sizes (42 for each condition) are not high enough for a reliable t-test. To evaluate the differences in the largest RT Goal distributions a paired Wilcoxon test was performed. In the Blank Screen condition, RT Goal distributions differed significantly between the two distraction conditions with p-value 0.001 ($w = 185$). But, in the word game condition, RT Goal distributions did not differ significantly ($w=333$, p-value = 0.14). However, a similar trend

in disruption is seen for the Menu condition. With a larger sample size, this difference may reach significance.

4.1.7 Variability Effects Due to Other Factors

We expect that, in principle, several other factors can influence RT Goal. The ones we identified are:

- **Task-subject attributes** like familiarity and pleasantness.
- **Demographic attributes** like age and gender.
- **Miscellaneous attributes** like screen time per day, time-of-day (when the participant did the experiment), task encoding time.

Analysing their influence is difficult and in some cases somewhat problematic.

For task-subject attributes the chosen tasks are from among those that subjects use in their daily lives so an effect on RT Goal is possible. So, we have collected data on two such attributes - familiarity, pleasantness.

Trying to understand the effect of these factors on RT Goal is difficult since for an attribute and ordinal level we have to aggregate RT Goal for different task-subject combinations so the analysis is somewhat problematic because subjective ratings are unlikely to be consistent across subject-task combinations.

4.1.8 Effect of Familiarity on RT Goal

The Ordinal Variable Familiarity has the group sizes given below. All the familiarity groups had almost equal representation ($\approx 25\%$ each) of interruption and distraction conditions. Trials under Loss of Content category were excluded.

Familiarity Group	1(Least)	2	3(Neutral)	4	5(Most)
Group Size	175	142	204	274	778
Mean of RT Goal (in seconds)	7.26	7.25	5.72	5.87	5.83
Variance of RT Goal (in seconds ²)	41.9	49.3	26.4	23.1	23.6

To test for the effect of familiarity on RT Goal, Kruskal-Wallis test was performed as variances and sample sizes are unequal. A significant difference in groups was found, $H(4) = 14.5$, $p = \mathbf{0.006^{**}}$

4.1.9 Effect of Pleasantness on RT Goal

The Ordinal Variable Pleasantness was observed to have the below group sizes. Pleasantness groups also had equal representation ($\approx 25\%$ each) of interruption and distraction categories. Trials under Loss of Content category were excluded.

Pleasantness Group	1(Least)	2	3(Neutral)	4	5(Most)
Group Size	196	157	361	355	504
Mean of RT Goal (in seconds)	7.51	6.30	6.31	5.64	5.69
Variance of RT Goal (in seconds ²)	47.2	42.6	29.7	19.8	21.1

Similar to the analysis of Familiarity, to test for the effect of pleasantness on RT Goal, Kruskal-Wallis test was performed as variances and sample sizes are unequal. A significant difference in groups was found, $H(4) = 17.7$, $p = \mathbf{0.001^{**}}$

4.1.10 Effect of RT Encode on RT Goal

RT Encode was tested for correlation with RT Goal under presence and absence of Interruption separately with mean values for each subject. The upper limit of RT Encode was capped at 20 seconds to eliminate any outliers as there was no real reason for a participant to take longer than 20 seconds to encode a given task. The plots and results are shown below.

Under the presence of the Interruption Task, a weak negative correlation (-0.19 r) of RT Encode was observed which was not significant for the chosen sample size (42 subjects).

Without the Interruption Task, no correlation was observed.

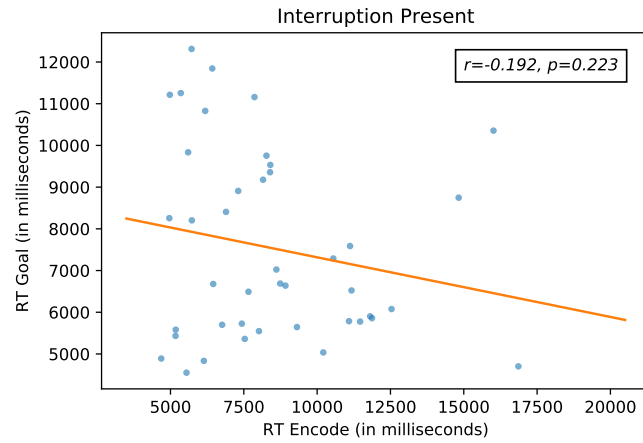


FIGURE 4.3: Correlation of RT Encode with RT Goal - Interruption Present

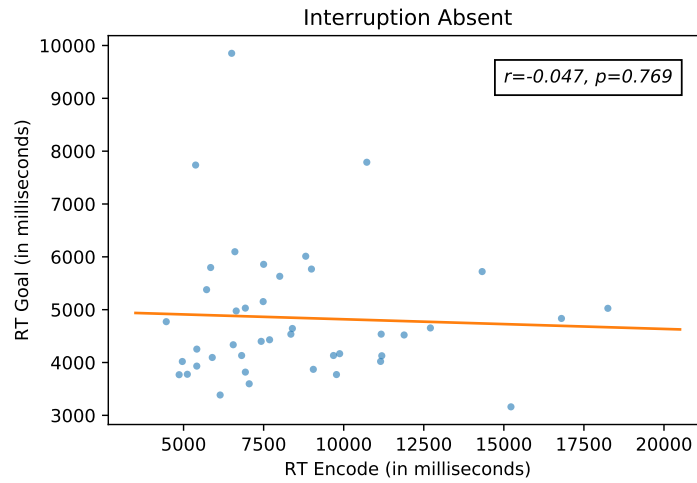


FIGURE 4.4: Correlation of RT Encode with RT Goal - Interruption Absent

4.1.11 Demographic Effects

4.1.11.1 Effect of Gender on RT Goal

Out of 42 participants, 25 were male and 17 were female. To test for any gender differences on RT Goal, Welch's t-test (unequal samples or variance, 2 classes) was performed. No significant difference was found.

T	dof	tail	p-val	CI95%	Cohen-d	BF10	power
0.452	1529.9	two-sided	0.651	{-395.5, 632.2}	0.022	0.062	0.073

4.1.11.2 Effect of Age on RT Goal

All participants were aged between 20 and 38 years. To test for any dependency between RT Goal and Age, a correlation test was performed. No correlation ($r=-0.008$) was found.

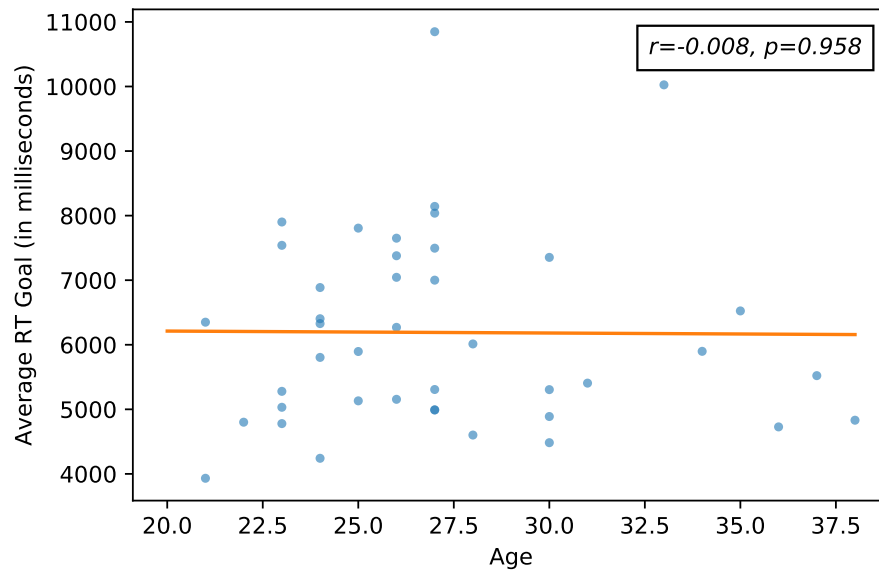


FIGURE 4.5: Age Correlation with RT Goal

4.1.11.3 Effect of Screen Time on RT Goal

To test the effect of daily screen time (smartphone) on RT Goal, Kruskal-Wallis test (>2 groups, unequal variances and sample sizes) was performed. Group Sizes and test results are shown below.

Screen Time Group	<3 hours	3-4 hours	4-5 hours	>5 hours
Group Size	6	18	7	11
Mean of RT Goal (in sec)	6.4	5.6	5.6	7.2
Variance of RT Goal (in sec ²)	32.5	21.1	25.0	39.5

A significant difference in Screen Time groups was found, $H(3) = 39.5$, $p < 0.001^{***}$. However, the sample size of three groups are way below the recommended sample size (15) for performing the Kruskal-Wallis test and no linear trend is observed on the group-wise means of RT Goal. Hence these results are inconclusive.

4.1.11.4 Effect of Experiment Start-time on RT Goal

Experiment start time i.e., time-of-day the experiment was started (in 24Hr format) was tested for linear correlation with RT Goal through a correlation test. No significant correlation was observed. The plot is shown below.

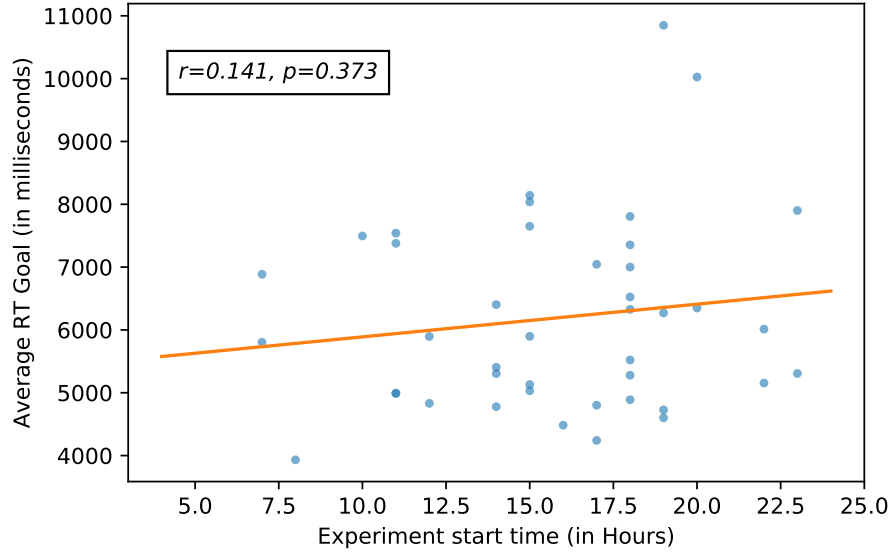


FIGURE 4.6: Experiment Start Time on Average Goal

4.2 Discussion

The experiment results show that Interruption has a significant effect on RT Goal, and Distraction has no effect, both in terms of differences in means [Section 4.1.3] and the number of Goal Slippage and Loss of Content instances [Section 4.1.1] between conditions. Compared with an assumed baseline or the ideal condition, every condition had a significantly different trial count [Table 4.3] suggesting that disruption is associated with delayed task execution. Although distraction conditions did not differ in RT Goal means, distributions were observed to be different, which prompted us to do further analysis. Under the Blank Screen condition, distraction conditions were found to have significantly different variances. As outliers or right extreme data of the Complete Retention category capture the trials with the most disruption, we have analysed the subject-wise right extreme RT Goal value differences between the two distraction conditions and found a significant difference [Section 4.1.6] under the blank screen condition. Although we could not find the same result under the word game condition, we see a similar trend in data which may

have led to significance with a larger sample. This implies that Menugrid favors some tasks or apps but at the same time hinders the selection of some others, whereas ‘Search’ has the most RT Goal values present around the mode or peak value. Later in this section, we bring out the practical significance for this difference and recommend that interface designs need to eliminate distractions whenever possible. We label the menugrid variant of the task as ‘recognition’ and the search variant as ‘recall’ based on the different ways of retrieving information during the Navigate stage.

Task variable factors, ‘Familiarity’ and ‘Pleasantness’ were found to have significantly different RT Goal groups suggesting that familiar and inherently pleasant tasks are easier to recollect in general. This variability is noticeable (as observed in the plots below) in the presence of interruption, clearly suggesting that interruption is disruptive and that additional search time is needed to recollect unfamiliar and/or unpleasant tasks. We are unable to gauge the interaction effect through a statistical test due to the large variability in the sample sizes of different Familiarity and Pleasantness groups.

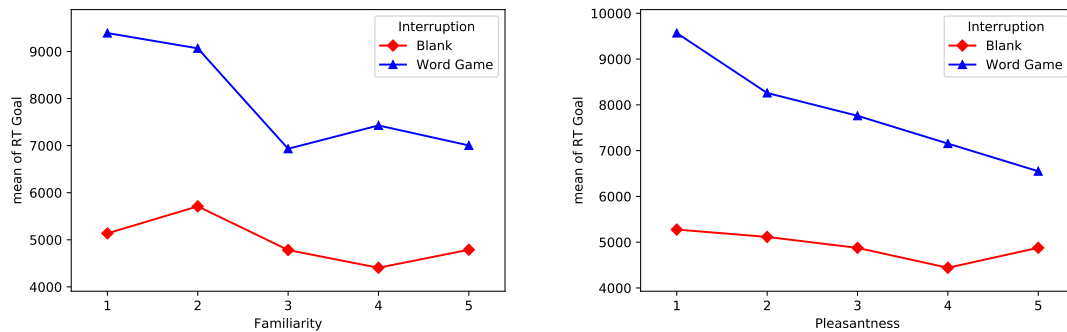


FIGURE 4.7: Interaction plots of Familiarity and Pleasantness (Levels ranging from 1-Least to 5-Most) on RT Goal.

A weak non-significant correlation was observed between RT Encode and RT Goal under the Word Game condition. Demographic attributes and Miscellaneous attributes were either not found to be affecting RT Goal or are inconclusive.

4.2.1 The practical relevance of 1 in 20 goal slippage while interacting with smartphones

From our experiment, we observe 1 goal slippage out of 200 tasks when the interruption is absent and 1 goal slippage out of 20 tasks when the interruption is present. The duration of the interruption task was set to just 18 seconds due to experimental constraints whereas

real interruption tasks on smartphones may last much longer and even multiply. These interruptions can arise from notifications that include text messages, phone calls, emails, workflow reminders, social media updates, and so on. A previous unfinished task could be another potential interruption while opening the device for accomplishing a new task. Some users might have a usage pattern of opening a particular set of apps (one or more) upon opening the device or a workflow that they develop over time with repeated usage. All such intrusions that the user engages in without performing the primary task fall under interruptions.

In drawing a parallel with the real-world setting, we would like to bring out another nuance in a successful recall that could not be incorporated in our experiment. After finishing the interruption task, especially a long one, it is necessary for one to have a ‘blank-in-the-mind’[15] moment, a meta-cognitive experience of hollowness that something that was unfinished must be recalled. Without that, the user might not even make an attempt to recall the primary task, as captured by studies on interruptions in the workplace situation[34][26]. However, in our experiment, we always cut off the interruption task after the duration mark and move to the Navigate phase where users are deliberately asked to recall the target app.

According to a 2018 study[25], the average user reaches out to their phone 58 times and spends 3 hours and 15 minutes each day. Out of the 58 sessions, some can be interruption-driven wherein there is no goal at hand for the user. And, in some sessions, there can be more than one goal that is intended to be accomplished. It is hard to know the exact count of task-driven sessions in a day but it is reasonable to conservatively predict that an average user faces one goal slippage in a day if not more while initiating an interaction with their device. Accumulating the occurrences of goal slippages over time and multiplying it with the billions of smartphone users worldwide, the scale of the problem can be large.

4.2.2 The practical relevance of the difference in tail and RT Goal variance between Recognition and Recall

It is important to set up the context for this argument with the current heuristic principle of Graphical User Interface(GUI) design. In UI design and development, empirical evaluation methods of experimenting with interface design choices using test users are not very prevalent in industry due to lack of time, expertise, or an inclination to follow tradition[32]. An alternative, heuristic driven, informal way of usability analysis is prevalent where opinion based judgements were made simply by looking at the interface. Nielsen and Molich

systematized this way of analysis and proposed Heuristic Evaluation[32]. Developed as a discount usability engineering method, ten heuristic principles were suggested for better usability.

‘Recognition rather than Recall’ is one of the 10 heuristics proposed[31]. According to Nielsen’s synthesis of usability principles[30], ‘See-and-point instead of remember-and-type’ is the most listed phenomenon within this specific heuristic. This phenomenon indicates the preference for the see-and-point interface choices. To our knowledge, there is no study that argues against this preference. It appears that this usability heuristic has dictated the design of interfaces to date.

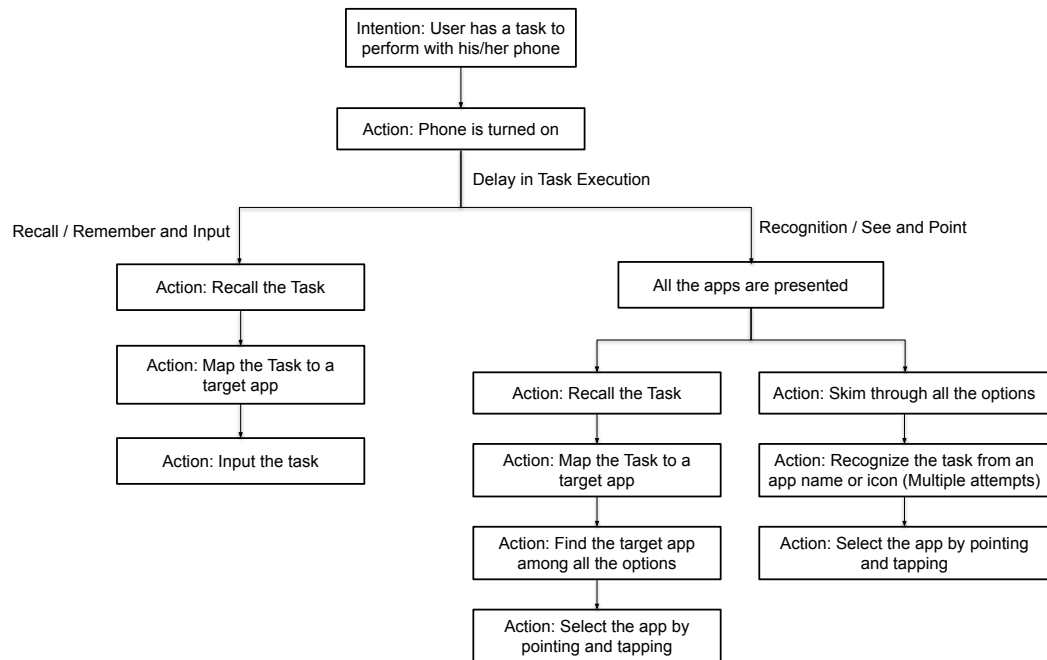


FIGURE 4.8: Recognition vs Recall

It is important to note that the comparison between Recognition and Recall here is not a direct representation of the difference in cognitive phenomena but rather a mere high-level differentiator between interaction modes. In practical scenarios, see-and-point interaction does not necessarily limit to recognition. One may carry out the interaction by recalling and then recognizing. The design principle is not attending to these nuances and is only making a high level distinction. A typical user flow through each of these interaction modes is depicted in the above diagram.

Usability refers to ease-of-use of a product and ease-of-use is defined comprehensively to include five dimensions - the 5Es of usability[36][37]. The 5Es are Effectiveness, Efficiency,

Engagement, Error Tolerance and Easy to Learn. Effectiveness is the completeness and accuracy with which users achieve specified goals, and Efficiency is the speed with which users can complete the tasks for which they use the product. For the current context of smartphone usability, effectiveness becomes the top dimension followed by efficiency.

Through our experiment we were able to gauge both these aspects in terms of goal retention and time taken to reach the goal (RT Goal). Going by the heuristic principle, we should have observed a clear preference towards the Recognition variant. The experiment's recognition variant is clearly advantageous on the surface. The visual cue is made available in an app icon that is familiar to users, and the app names are presented in alphabetical order. With one click on the app icon, the participants could navigate to their target app. For the Recall variant, participants had to first click on the search bar, start typing the letters of the recalled app icon, look for the target app from suggestions, and then click on it. The number of clicks is more than the Recognition variant and there is no visual cue in case they forgot the task.

We did not observe any significant difference between Recognition and Recall in the assigned trails of Goal Retention categories given the 30-second duration for recall, nor did we observe any significant difference between the means of RT Goal. However, the wider spread of RT Goal during Recognition is significant under Blank Screen condition. And specifically, when right-end tails are selectively analysed, Recall has a clear preference over Recognition. The right-end tail indicates clear disruption if not complete goal slippage. Suppose we limit the recall time to 12 or 15 seconds and assign everything beyond that to goal slippage; we may have seen effectiveness in favor of Recall and efficiency in favor of Recognition. Note that effectiveness is more vital than efficiency. With a stricter recall duration and/or a stronger interruption task, we predict that there will be a clear preference for Recall over Recognition from the main result itself.

4.2.3 How ‘Recognition rather than Recall’ is harming Digital Wellbeing?

Recognition invites the user to scan through all the available options and encourages engagement with the device. This is a great opportunity for businesses to grab the user's attention and propagate interruption-driven processing also known as hooks. As this variant is the default and in fact the only variant available at a system level or an application level, it is inevitable that users end up spending much more time on their devices than

intended. The phenomenon of Goal Slippage further argues against it and questions the choice of the Recognition variant for the interface in the first place.

If one aims to design for digital wellbeing, the Recall variant should be preferred over Recognition. Recall or Search UI Components can be developed to be smarter and intelligent where the device is voice assistive for efficient interaction and is able to understand the target app with a relevant keyword and possibly even interpret the task within the app and reach the page within the app. Recall can then challenge the Recognition variant in efficiency. The Google Search engine homepage is a good example of the Recall variant. We envision an UI System on smartphones that functions in a similar manner, free of interruptions and distractions.

4.2.3.1 Conditions for 'Recall rather than Recognition'

Synergy between the system and the user is necessary for a successful interaction based on Recall. A user needs to declare her/his intention/task and the system should comprehend what should be done. The conditions are:

1. The user is familiar with the functionalities of the system.
2. The user is able to articulate his/her intention/task at hand in a way that is comprehensible by the system.
3. The task is unambiguous for the system or the system is able to disambiguate the articulated task
4. The user is certain that the system can comprehend the recalled command as intended.

Chapter 5

Design for ‘Focus Mode’

In this chapter, we present our design approach in attempting to Design for Digital Well-being and propose a Design System that re-imagines the smartphone launch experience.

5.1 Survey

We have conducted a user survey(n=35) on smartphone usage for the age group 20 to 38 years. Participants were either students or working professionals. Here are the results of the survey

Control (6 items)	1 (Not at all)	2	3	4	5 (Always)
I tend to lose track of time when I’m on my phone.	3	7	6	13	6
I feel the need to check my phone right away if it vibrates.	4	6	5	12	8
I spend more time on social media than I’d like.	7	4	9	12	3
I’ll stay on my phone instead of going to sleep when I intend to.	6	6	7	13	3
I feel like I’m missing something important at work if I don’t check my phone.	5	6	12	10	2
I feel overwhelmed by the amount of content that I have to attend.	7	7	9	10	2

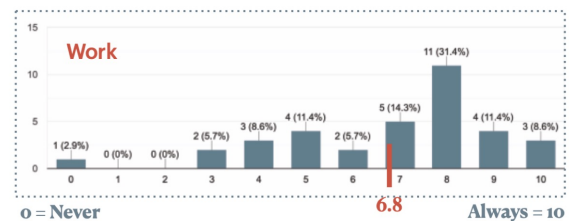
Dependency	1 (Not at all)	2	3	4	5 (Always)
“If I have a question I reach for my phone for an immediate answer.” How much do you agree with this statement?	1	2	2	10	20

Activity	Less than an hour	1 to 3 hours	3 to 5 hours	More than 5 hours
How much time do you think you spend on your smartphone in a day?	2	13	11	9

(On a scale of 0-10)

How much do you rely on your smartphone for Work or Goal-related activities?

Example: Study / Finance / Calling / Shopping with an actual need



(On a scale of 0-10)

How much do you rely on your smartphone for Leisure or Non-goal-related activities?

Example: Social Media / Movies / Gaming / Shopping without an actual need

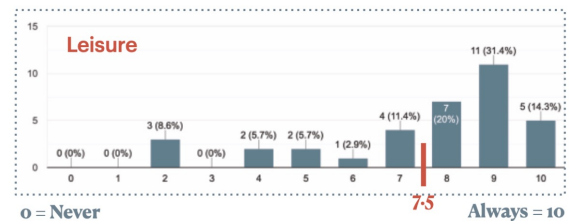


FIGURE 5.1: Activity - Work and Leisure

Goal Slippage	Yes	No	Maybe
While interacting with your smartphone with a specific goal, do you ever tend to forget your goal?	17	12	6

From the survey results, many concerns with smartphone usage and the need for digital well-being become evident. Users are aware of their dependency and a lack of control over their usage. We also see that smartphones are being considered one-for-all solutions, where they are the go-to devices for both Work and Leisure.

Towards the end of the survey, we attempted to explain our focus area, ‘Goal Slippage,’ with an example, ‘You intend to set up an alarm, but a message from a friend interrupts you. And, now you can’t even recollect your goal (setting up the alarm)’. Participants

were able to grasp the idea, and some of them (7 users) shared their experiences in the optional description field.

We have conducted semi-structured interviews with these 7 users to understand their usage and empathize with their context. The below scenarios, supplemented with user personas, were created to translate the actual instances of goal slippage.

5.2 User Scenarios

5.2.1 Persona and Scenario - 1

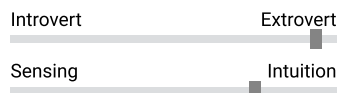
Manohar

26 years | Male | Single
Frontend Developer | Bangalore

- Manohar is set out to establish himself outside of his office work.
- Manohar is a social butterfly. He always wants to stay connected with his friends.

Device	Time on Device
Laptop (Work)	7 hours
Samsung J6 (Personal)	4 hours

Manohar is working as a front-end developer at a startup company. His family is dependent on him. However, he doesn't intend his work role to be his only recognition or career path. He is good at storytelling and writing. He showcases his creative skills through his short articles and opinion pieces on social media platforms. He is now in trials to author a book and works on it whenever he can.



He has over 1000 connections on social media whom he has met in various stages of life while in school, college and work. He is active in multiple media groups with his close friends and is connected to his friends via social media platforms throughout the day, even during work.

More often than not, I would open an app that is different from the one I intend to use in the first place before unlocking the screen.

Tuesday 4PM, work-desk

1. Manohar finished his work early and his daily stand-up meeting won't start until 30minutes.
2. He thinks it is best to make use of this time by continuing his story for his upcoming book and reaches for his phone to open the draft from his note-taking app.
3. Upon unlocking the device, he sees a notification from a group chat mentioning that he is tagged on a couple of messages. (Group: College buddies, Members: 12, Active: 3)

4. Manohar opens the group chat and finds a meme shared by close friend. It is on his favorite cricket player. A couple of friends were laughing and making fun.
5. Manohar initially laughs along but tries to make a point by pointing to some of the moments where his favorite player played a key role. A friend of his opposes.
6. He then starts searching for the exact details and shares the video snippets. They argue with stickers and punch lines. Few more friends join and the conversation grows.
7. It's 4:28PM and Manohar receives a reminder notification from his work laptop.
8. He leaves the conversation in between and prepares to join the daily stand-up to share his work updates.

5.2.2 Persona and Scenario - 2

Archana

23 years | Female | Single
Grad Student | Delhi

- Archana is highly motivated to take a job after her course.
- Archana likes to perform all tasks from her smartphone.

Device	Time on Device
Old Laptop (Study)	2 hours
iphone SE 2 (Mixed)	6 hours

Archana is a final year master's student of agriculture in a premier university. Archana strongly desires to secure a job by the time she finishes her course.

She attends classes virtually from her hostel. She receives class updates from her teachers and batchmates on a messenger platform. She prefers to attend lectures from her smartphone as her laptop is quite old, which she uses only for assignment submissions.



My fingers automatically reach for WhatsApp Icon whenever I open my phone.

University Library, Books on Desk

1. Archana is reading a book on floriculture and encounters a new word.
2. She reaches out to her mobile phone to look for its meaning.
3. Upon unlocking the device, she sees the social media app icons on her home screen.

4. It's her routine to check these apps as these are important to her. As she turns off all her social media notifications, there is always something new.
5. She first looks for any messages on the college group chat. She reads them nothing seems prominent.
6. She checks the responses on her recent post from the group page that she owns. She responds to some.
7. She then closes the device and turns towards her book wondering why she had opened her phone.

5.2.3 Persona and Scenario - 3

Ashutosh

30 years | Male | Single
Ph.D Scholar | Kanpur

- Ashutosh is ready to spend any number of hours on his research work.
- Ashutosh believes in overall development as a person.

Device	Time on Device
Dell XPS (Mixed)	9 hours
Moto G6 (Personal)	3 hours

Ashutosh is pursuing research as a Ph.D. student. He is passionate about his work and the impact it can have on society. He works in a research lab with 4 other fellow researchers and leads them. He is exploring opportunities and evaluating the feasibility of his ideas. His team members reach out to him for collaboration.



Ashutosh is health-conscious and exercises daily. In his leisure, he watches documentaries and videos from bloggers. He is subscribed to many channels that produce quality content on social media platforms.

I always find myself lost into a vortex when I'm in Youtube.

Research Lab, Working on Laptop & Monitor

1. Ashutosh is detailing out his new idea. He feels that there should be some existing material on the idea that he is exploring and opens a video sharing platform.
2. Upon opening the website. he first attends to the news-feed.
3. He finds that there is a new video released from a vlogger that he adores.
4. Intrigued by the title and thumbnail image he clicks on it.

5. While watching it he also checks the comments made by viewers.
6. He thoroughly enjoys it and appreciates the video with a ‘Like’.
7. After the video, another recently published video from the same vlogger is set on auto-play. Ashutosh is interested in this as well and let’s the next video roll.
8. He then finds a similar video on the same concept from the suggested videos and clicks on it to see how the other vlogger has presented it.
9. While watching this he feels guilty for spending more time than intended, pauses the video and looks away trying to recollect what made him open the video sharing platform

5.2.4 Persona and Scenario - 4

Geetha

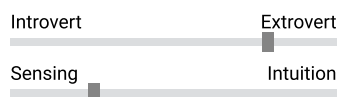
35 years | Female | Married
Content Writer | Hyderabad

- Geetha is always in search of novels and fiction series.
- She is genuinely interested in the well-being of her friends and family.

Device	Time on Device
Thinkpad Laptop (Work)	5 hours
iPad (Personal)	4 hours

Geetha is working as a consultant for content writing and is outsourced to a client. She receives weekly assignments to work on. She prefers to work in her own time and doesn't have any strict work hours.

Geetha is an avid reader. You'll find her surrounded by books and an iPad any time of the day. She doesn't have a lot of friends, but the ones she is friends with are very close to her. She loves spending time with them



I think I have messaged/replied to a message, as I move on to my reading.

Home 2PM, Lunch

1. Geetha is having her lunch on a Thursday.
2. She has been long planning to invite her colleagues for lunch on a weekend and this Saturday seems to be just right. She tells herself ‘Let’s drop an invite in the office channel’

3. She finishes her lunch, wipes the table and puts away the dishes in the dish corner
4. (2:15PM) She lays on couch and grabs her iPad.
5. Upon unlocking the iPad, her recently opened app shows up. It’s the novel which she started yesterday, She is reminded how intrigued she was after the first 2 chapters.
6. She proceeds to read where she has left off.
7. (2:45PM) She finishes one more chapter. She is feeling sleepy as she is used to resting for an hour in the afternoon.
8. She grabs her pillow and leaves her iPad aside.

5.3 Analysis

The four scenarios portray various kinds of goal slippages in real-life. In all cases, the interface of the phone is driving the interaction by drawing users to interact with what is presented and guiding them into the vortex(in Ashutosh’ terms, persona 3).

	Scenario 1	Scenario 2	Scenario 3	Scenario 4
Distractions Level	System-Level	System-Level	Application-Level	System-Level
Distraction that got converted into an Interruption	Notification of Group Chat	App Icon	Suggested Video Thumbnail	Recently opened application
Attempt to Recall Primary Goal	Never	After closing the device	Towards the end of the session	Never
Theoretical Reason	Work-Leisure Mixup	Recognition rather than Recall	Interruption Driven UI	Goal Assumption

Current smartphone launchers promote interruption driven interaction through a variety of sources of distractions such as

- Recently viewed application
- Notifications
- App icons (and Notification Dots)

- Widgets

Application home screens hold distractions such as

- Unread content (Messages/Mails)
- Suggested content on Timeline
- Notifications
- Update prompts (Dialogs/Popup windows)
- Action prompts with notification dots

For task-driven interaction, the launch experience must be free from the above distractions. The device can prompt the user to ask for their immediate/primary goal, thus initiating their intended action. Here’s the generalized desired flow for a task-driven smartphone launch interaction.

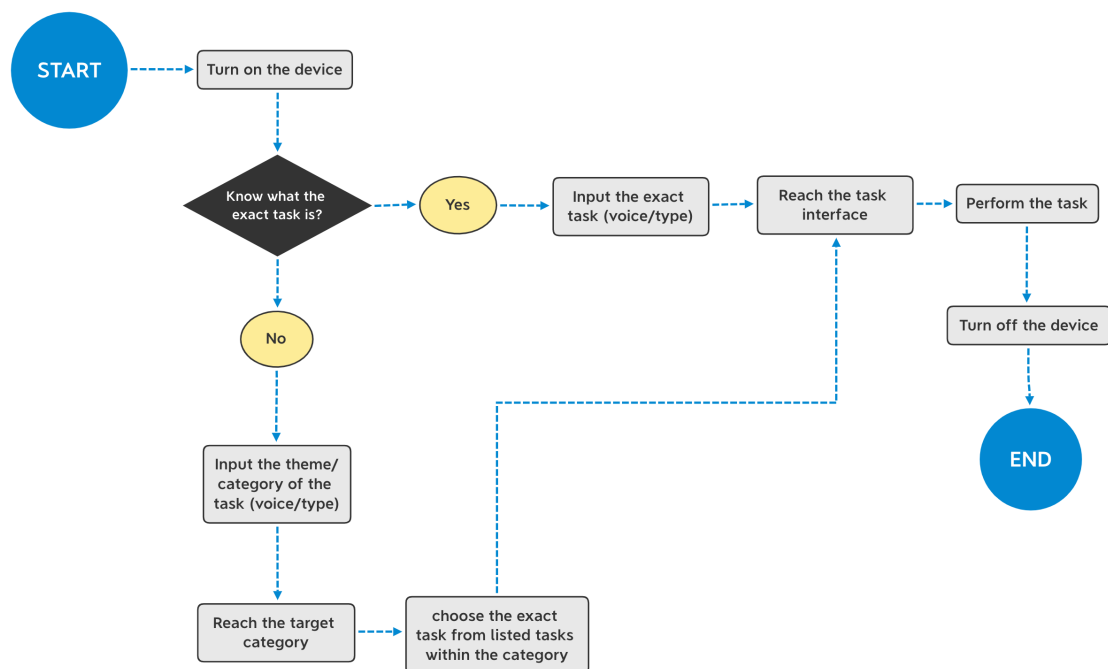


FIGURE 5.2: Generalized Flowchart for Task-driven Interaction

Focusing on the major area in the flowchart, which is for the condition when an exact task is known, we performed scenario mapping to capture possible approaches to carrying out the specific interaction steps. We consider questions for each step and ideas towards a potential solution at this stage.

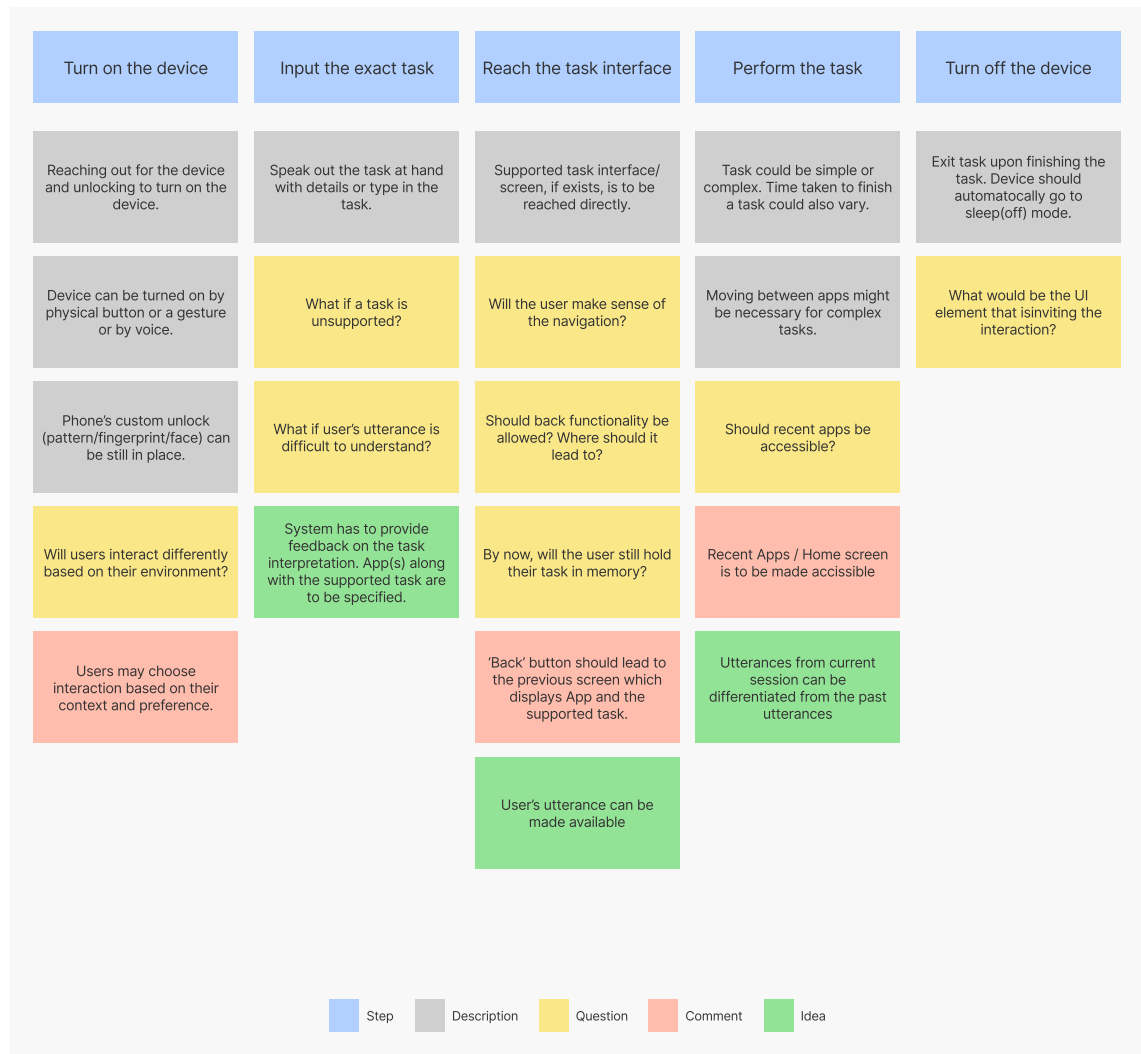


FIGURE 5.3: Scenario Mapping for Smartphone Launch Experience

5.4 Design Principles

During the analysis, certain unifying elements were found to be guiding our design solution. Here we establish them as design principles.

5.4.1 Immediate Task Execution through Recall

Always ask what the user wants before making any recommendations. With evidence from our study, we propose our first principle that argues against the traditional design heuristic for usability, ‘Recognition rather than recall,’ to mitigate distractions during an interaction and prevent goal slippage. Recall probes for immediate task execution for the user.

5.4.2 Tasks, not apps

Let the user interact with the task at hand. Users’ objective is to complete a task via their smartphone, and an app is only a medium to accomplish it. The task-driven interface should allow direct interaction without forcing users to translate their tasks into an app’s functionality or architecture. App home screen is an additional layer in the interaction holding application-level distractions. Whenever feasible, navigation must eliminate them, allowing for direct interaction.

5.4.3 Semantic Categorization

System to decode tasks based on the meaning. We rely on the semantic categorization of the user’s utterance to direct them onto their target page, thereby eliminating intermediate steps that otherwise necessitate the user to recollect the app’s name or recognize its icon, navigate through the app’s architecture.

5.4.4 Exit points

Provide clear exit points after task completion. Engaging UI is so much about conversion rates that having additional content/tasks after accomplishing a primary task is the norm for websites and applications. To actively cut off from an engaging interface, we employ exit points as a principle to mark task completion and encourage users to get back to their activity.

5.4.5 Limit Passive usage

Don’t let the user stare at their screen without a purpose. Smartphone interactions have developed significantly to keep users glued to the screens for hours. While bringing in the best practices from the current UI, we restrain using interactions that support mindless or passive use and adopt this mindset of limiting passive usage.

5.4.6 Promote Voice Interaction

Encourage active usage with voice interaction. Voice interaction enables immersive and efficient interaction for task-based interaction.

5.5 Focus Mode - Design Concept

Applying the stated design principles, we envision a smartphone launcher with minimal interaction elements. As the sole purpose of the design is to support the user’s focus on their activity outside of smartphone interaction, we call the launcher ‘Focus Mode.’ In this section, we present the concept for Focus Mode through wireframes.

The launcher’s home screen accommodates just two elements, one for text input and the other for voice input, to receive utterance of the user’s primary task as is.

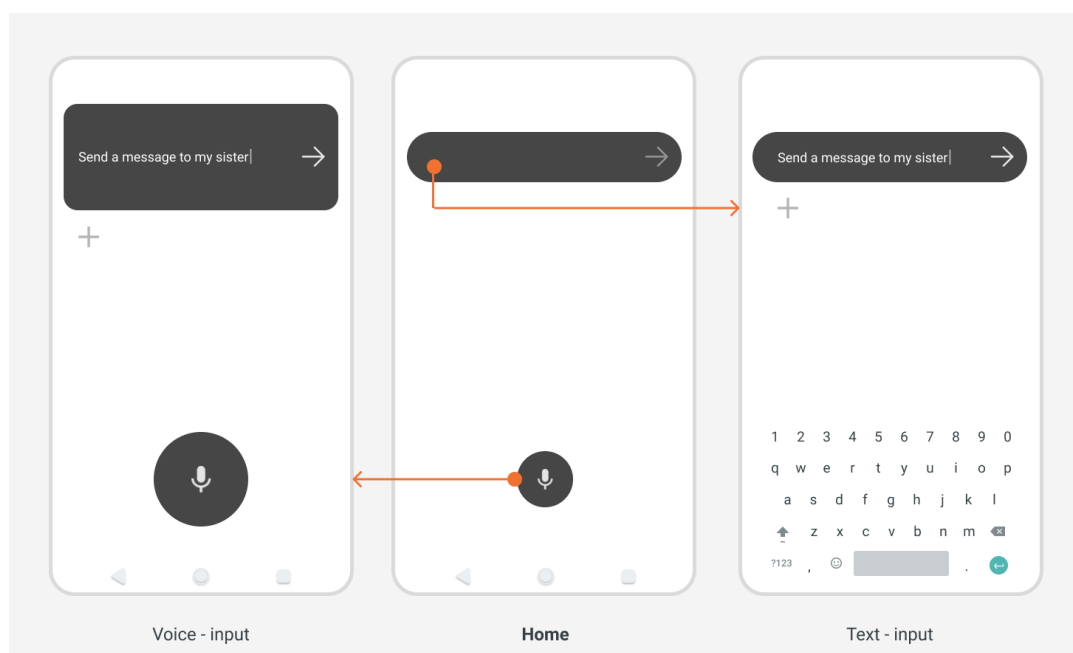


FIGURE 5.4: Launcher Interaction

The user’s utterance from the free input can be analyzed using semantic categorization to extract App name and an appropriate task (if it exists) from the utterance. The table below presents some utterances with possible contestants for App & Task within the app. When both are known, the launcher can navigate the user directly to the target screen.

Utterance	App	Contestant for App	Task	Contestant for Task
Where is my Amazon order?	Known	Amazon	Known	Track Order
Gmail	Known	GMail	Default	Homepage
Check my bank statement	Known	Banking App	Unclear	Statement(3), Homepage
Send a message to my sister	Unclear	Messaging Apps(3)	Known	Conversation
Open my assignment	Unclear	Files(3)	Default	Homepage
What is the live score	Default	Search Engine	Known	Search Result

If an utterance doesn’t clarify the target App, the launcher can either assign it to the default app or prompt the user to rephrase their utterance with a target app name or class. When there are multiple contestants for App/Task, a user can pick their target from the provided options. For Task, a default option of Homepage can always be made available to cater for incorrect suggestions. The suggested navigation is shared below.

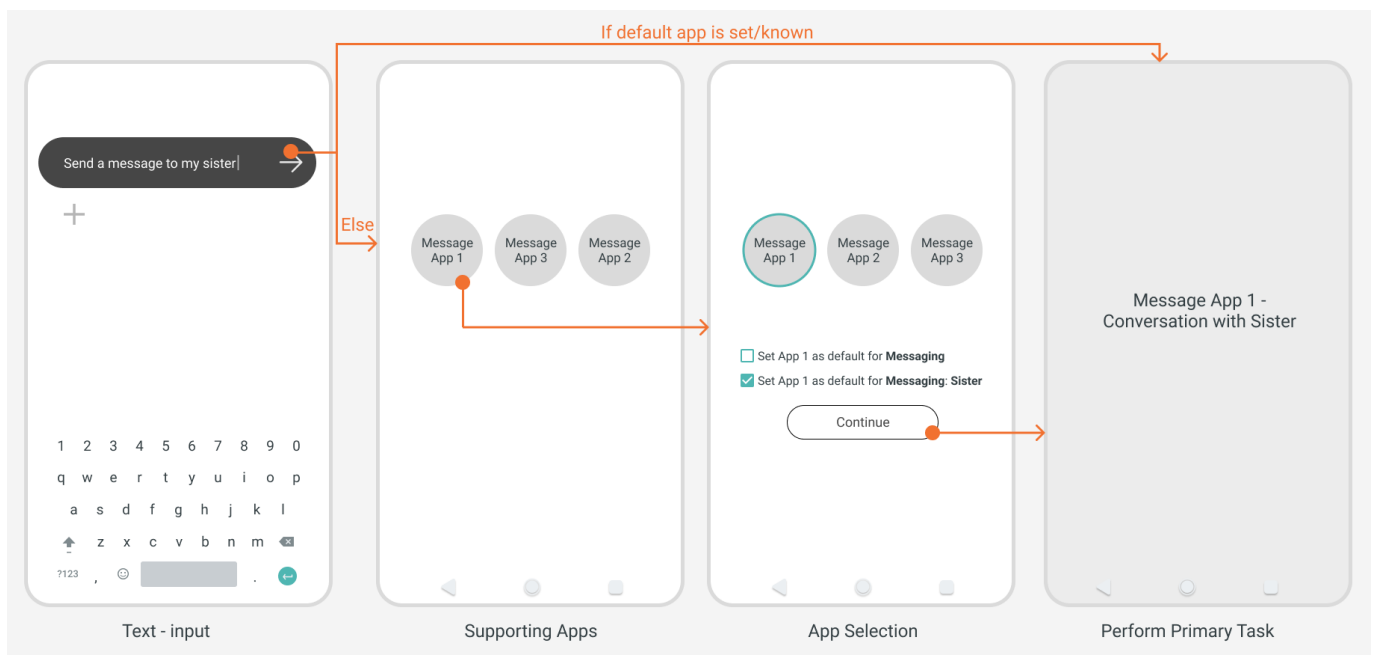


FIGURE 5.5: Navigation for App Selection

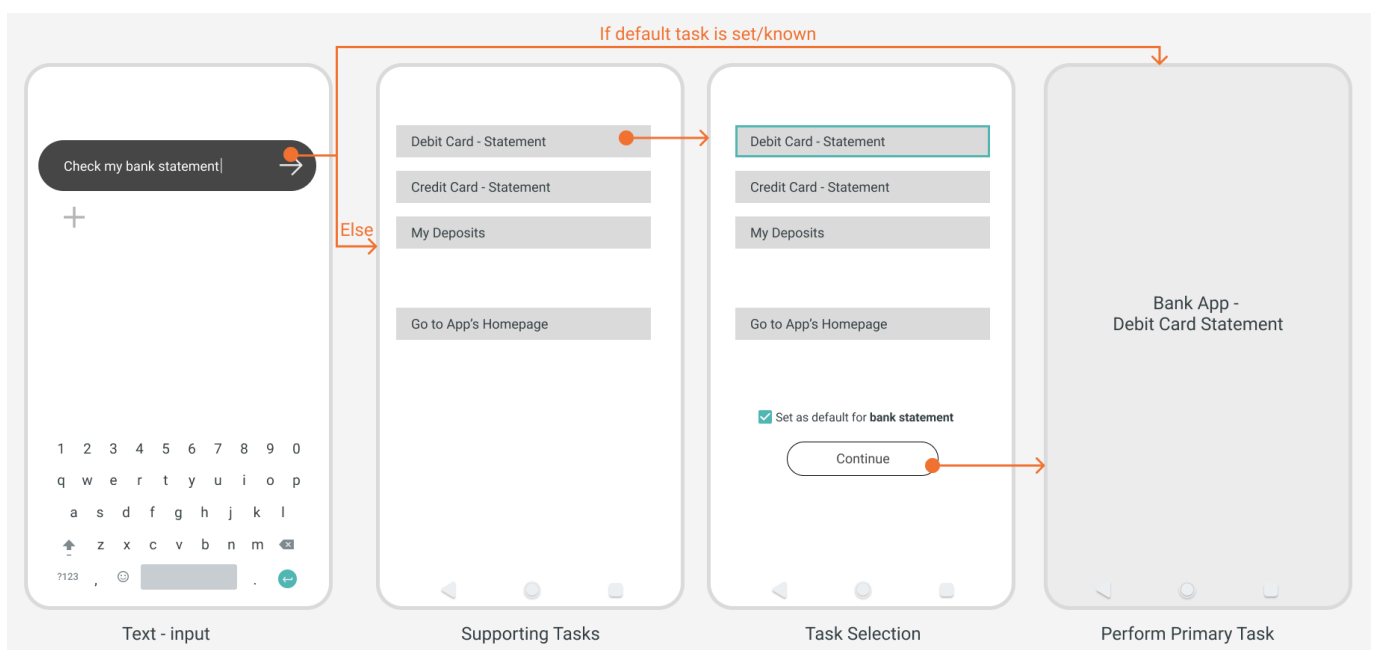


FIGURE 5.6: Navigation for Task Selection

A user can have more than one primary task in mind while launching the device. The launcher can support noting down multiple tasks(reference shared below), and they are erased from the home screen as the user accomplishes them. A task is considered accomplished when a user navigates to the home screen from their primary task screen. Thus, a specific screen of a target app is accessed, and the user is provided with clear exit points. Instead of the traditional recent apps screen, recent tasks are made available with a timestamp, as shown below. They can be accessed with an utterance similar to ‘Open Recent Tasks’.

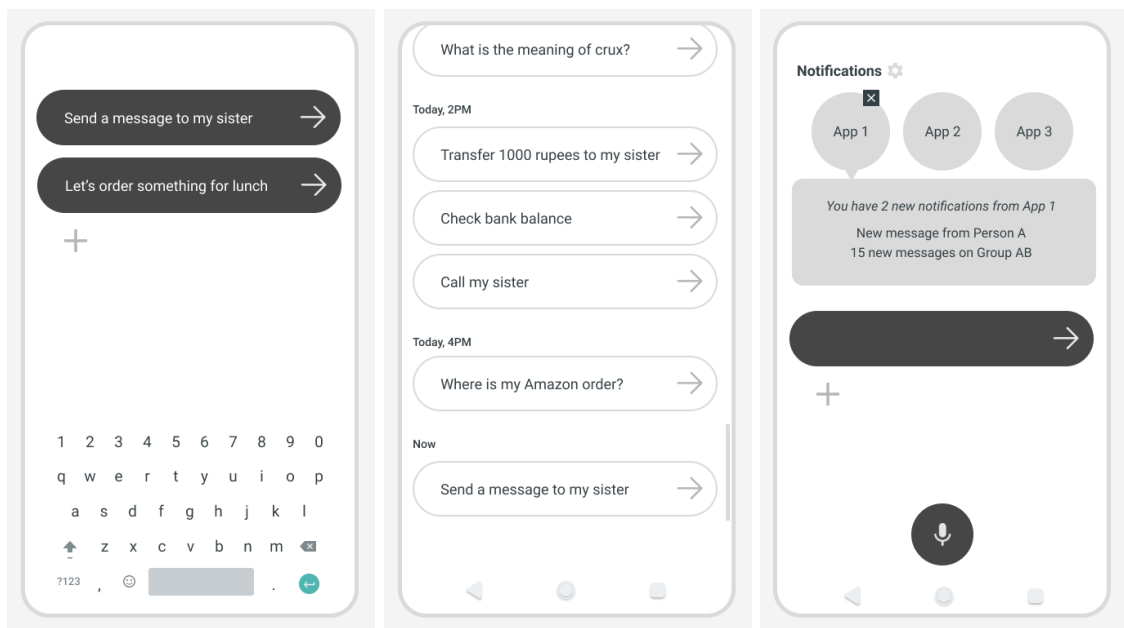


FIGURE 5.7: Supporting Multiple Tasks on Home Screen, Displaying Recent Tasks and Notifications

Utterance	App	Contestant for App	Task	Contestant for Task
Recent App	Known	Most recent app	Known	Associated Task page
Recent Apps (or tasks)	Display recent tasks			
Show my Notifications	Display notifications			

Chapter 6

Conclusion

With our experiment, we found that Interruption has a significant effect on RT Goal, and Distraction has no effect, both in terms of differences in means [Section 4.1.3] and number of assigned Goal Slippage and Loss of Content instances [Section 4.1.1] between conditions. Compared with an assumed baseline for immediate recall, every condition had a significantly different trial count [Table 4.3] suggesting that disruption is associated with delayed task execution. Under the Blank Screen condition, distraction conditions were found to have significantly different variances. While selectively analysing the subject-wise largest RT Goal values under complete retention between distraction conditions, significant difference was found for the blank screen conditions and a non-significant similar disruption trend was observed for the word game conditions. [Section 4.1.6] This indicates clear disruption from a MenuGrid if not complete goal slippage. We looked at the practical significance of this difference in variance and recommend that interface designs need to eliminate distractions whenever possible. With a stricter recall duration and/or a stronger interruption task, we predict that there will be a clear preference for Recall over Recognition through the main result which is two-way ANOVA. Task related factors, ‘Familiarity’ and ‘Pleasantness’ were found to differ significantly between groups suggesting that familiar and inherently pleasant tasks are easier to recollect in general.

We have proposed a Design System, Focus Mode, that re-imagines the smartphone launch experience, guided by the design principles: ‘Immediate task execution through Recall,’ ‘Tasks, not Apps,’ ‘Semantic Categorization,’ ‘Exit Points,’ ‘Limit Passive Usage,’ and ‘Promote Voice Interaction.’ The launcher consists of minimal interaction elements while supporting touch and voice input. The home screen is set up to receive utterances of the

user’s primary task. A specific screen of a target app can be accessed directly from the launcher home screen using semantic categorization of the user’s utterance.

6.1 Future Work

We believe that Focus Mode, if developed and put into use, will positively impact users’ digital wellbeing, covering aspects of passive usage prohibition, mitigating goal slippage, and retaining the user’s sense of control on device usage. Future work will encompass testing the design system against each of these wellbeing’ attributes.

Specific to goal slippage, further analysis and large-scale subjective reporting is useful in establishing the working memory concern in the design of digital applications.

Specific to the experiment, the baseline group (such as immediate recall in our case) should have been a separate condition while setting up the factorial design to eliminate any assumptions. As distraction variants, any additional (or new) interface options can be tested as an alternate to Menu or Search alternatives that have been used in the current study. We also recommend using multiple interruption durations whenever feasible to capture the associated variability in recall.

As one of our findings suggests that the inherent variability of the task affects its recall, we recommend exploring such variables in producing a comprehensive set of variables that represent the modern-day nature of tasks held in memory.

Appendix A

Primary Tasks

- Look for availability of the wishlisted item on Amazon
- Look for a smartphone under Rs 15000 on Amazon
- Look for any new movies in PVR on BookMyShow App
- Check seat numbers of your 4 tickets on BookMyShow App
- Set an alarm for tomorrow morning at 7 AM in the Clock App
- Start a timer for 15 minutes in the Clock App
- Book a train ticket on the IRCTC website on Google Chrome
- Delete your past 3 days browsing history from Google Chrome
- Upload your resume to the Documents folder in Google Drive
- Share your project folder with your teacher on Google Drive
- Go to your profile page and update your bio on Facebook
- Send a friend request to your friend Rohit on Facebook
- Check if the OTP was delivered to the Spam folder in Gmail
- Draft a mail to your colleague 'Bharat' on Gmail
- Search for the meaning of the word 'Idiosyncratic' on Google
- Check for the live score of a cricket match on Google

- Follow Hrithik Roshan's official page on Instagram
- Send birthday wishes to your friend on Instagram
- Rate the salon that you recently visited on Maps
- Find the neighboring countries of Belgium on Maps
- Check recent stories from your friends on Messenger App
- Join the group video call of school friends on Messenger App
- Watch the English movie 'The Irishman' on Netflix
- Look for crime fiction genre TV shows on Netflix
- Recharge your number for 10GB data on Paytm
- Add Rs 150 balance to your Wallet on Paytm
- Call your brother to know about his whereabouts from Phone App
- Check when the last call from your parent was made on Phone App
- Check if your android version is up to date on the Settings App
- Set your phone ringtone to default on Settings App
- Look for the latest tweets with the hashtag #Education on Twitter
- Tweet about a giveaway promo code on Twitter
- Book a cab to the railway station on Uber App
- Add 'credit card' as a new payment method in Uber App
- Share your hostel location with your sister on Whatsapp
- Check new messages from your family group on Whatsapp
- Watch the trailers of Christopher Nolan's movies on YouTube
- Check the latest video from 'T-Series' channel on YouTube
- Order a medium pizza from Dominos on Zomato app
- Look for a nearby restaurant with 4.5+ rating on Zomato app

Appendix B

Instructions

Welcome to the Experiment!

Instructions

There are 2 practice trials to familiarize you with the experiment.

In each trial, a task will be given to you followed by an interval or a word game. After that you will have to navigate to the target app within the experiment (via search or menu grid) and try to recall as much content of the task as possible. Take your time to properly memorize the task given to you.

Word Game: Make words by rearranging 3 or more letters of the given word. For example the word '*SILENT*' can be rearranged to make new words such as Sit, Lens, Ten, List, Listen,...

Encode | Hold / Game | Navigate | Recall

PRACTICE

DISCLAIMER: The data collected in the experiment will be kept confidential. You will be asked to provide your email ID in the next page. It is being collected solely for the communication purpose - to deliver your voucher after the successful completion of experiment.

FIGURE B.1: Instructions at the beginning of the experiment

Bibliography

- [1] Alboher, M. (2008). Fighting a war against distraction. *The New York Times*. Retrieved from <http://www.nytimes.com>. Last accessed 5 January 2022.
- [2] Altmann, E. M. and Trafton, J. G. (2002). Memory for goals: An activation-based model. *Cognitive science*, 26(1):39–83.
- [3] Boehm-Davis, D. A. and Remington, R. (2009). Reducing the disruptive effects of interruption: A cognitive framework for analysing the costs and benefits of intervention strategies. *Accident Analysis & Prevention*, 41(5):1124–1129.
- [4] Boneau, C. A. (1960). The effects of violations of assumptions underlying the t test. *Psychological bulletin*, 57(1):49.
- [5] Brandimonte, M. A., Einstein, G. O., and McDaniel, M. A. (2014). *Prospective memory: Theory and applications*. Psychology Press.
- [6] Brevers, D. and Noël, X. (2013). Pathological gambling and the loss of willpower: a neurocognitive perspective. *Socioaffective Neuroscience & Psychology*, 3(1):21592.
- [7] Brignull, H. (2018). Dark patterns. Retrieved from <https://www.darkpatterns.org/>. Last accessed 5 January 2022.
- [8] Brumby, D. P., Janssen, C. P., and Mark, G. (2019). How do interruptions affect productivity? In *Rethinking Productivity in Software Engineering*, pages 85–107. Springer.
- [9] Cecchinato, M. E., Rooksby, J., Hiniker, A., Munson, S., Lukoff, K., Ciolfi, L., Thieme, A., and Harrison, D. (2019). Designing for digital wellbeing: A research & practice agenda. In *Extended abstracts of the 2019 CHI conference on human factors in computing systems*, pages 1–8.
- [10] Clapp, W. C., Rubens, M. T., and Gazzaley, A. (2010). Mechanisms of working memory disruption by external interference. *Cerebral cortex*, 20(4):859–872.

- [11] Couffe, C. and Michael, G. A. (2017). Failures due to interruptions or distractions: A review and a new framework. *American journal of psychology*, 130(2):163–181.
- [12] Cowan, N. (2010). The magical mystery four: How is working memory capacity limited, and why? *Current directions in psychological science*, 19(1):51–57.
- [13] Dabbish, L., Mark, G., and González, V. M. (2011). Why do i keep interrupting myself? environment, habit and self-interruption. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, pages 3127–3130.
- [14] Deb, A. (2015). Phantom vibration and phantom ringing among mobile phone users: A systematic review of literature. *Asia-Pacific Psychiatry*, 7(3):231–239.
- [15] Efklides, A. and Touroutoglou, A. (2010). Prospective memory failure and the metacognitive experience of “blank in the mind”. In *Trends and prospects in metacognition research*, pages 105–126. Springer.
- [16] Elhai, J. D., Yang, H., and Montag, C. (2020). Fear of missing out (fomo): overview, theoretical underpinnings, and literature review on relations with severity of negative affectivity and problematic technology use. *Brazilian Journal of Psychiatry*, 43:203–209.
- [17] Eyal, N. (2014). *Hooked: How to build habit-forming products*. Penguin.
- [18] Flaxman, S., Goel, S., and Rao, J. M. (2016). Filter bubbles, echo chambers, and online news consumption. *Public opinion quarterly*, 80(S1):298–320.
- [19] Gray, C. M., Kou, Y., Battles, B., Hoggatt, J., and Toombs, A. L. (2018). The dark (patterns) side of ux design. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, pages 1–14.
- [20] Iqbal, S. T. and Horvitz, E. (2010). Notifications and awareness: a field study of alert usage and preferences. In *Proceedings of the 2010 ACM conference on Computer supported cooperative work*, pages 27–30.
- [21] Kahneman, D. (1973). *Attention and effort*, volume 1063. Citeseer.
- [22] Kokkonen, J. (2017). Sketching an elusive problem of distraction.
- [23] Lyngs, U. (2019). Putting self-control at the centre of digital wellbeing. In *2019 ACM CHI Conference on Human Factors in Computing Systems (Position Paper)*.
- [24] Lyngs, U., Lukoff, K., Slovak, P., Seymour, W., Webb, H., Jirotko, M., Zhao, J., Van Kleek, M., and Shadbolt, N. (2020). ‘i just want to hack myself to not get distracted’

- evaluating design interventions for self-control on facebook. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, pages 1–15.
- [25] MacKay, B. J. (2018). Screen time stats: How your phone impacts your workday – rescuetime. Last accessed 5 January 2022.
- [26] Mark, G., Gonzalez, V. M., and Harris, J. (2005). No task left behind? examining the nature of fragmented work. In *Proceedings of the SIGCHI conference on Human factors in computing systems*, pages 321–330.
- [27] Miller, G. A. (1956). The magical number seven, plus or minus two: Some limits on our capacity for processing information. *Psychological review*, 63(2):81.
- [28] Miyata, Y. and Norman, D. A. (1986). Psychological issues in support of multiple activities. *User centered system design: New perspectives on human-computer interaction*, pages 265–284.
- [29] Nadler, R. (2020). Understanding “zoom fatigue”: Theorizing spatial dynamics as third skins in computer-mediated communication. *Computers and Composition*, 58:102613.
- [30] Nielsen, J. (1994). Enhancing the explanatory power of usability heuristics. In *Proceedings of the SIGCHI conference on Human Factors in Computing Systems*, pages 152–158.
- [31] Nielsen, J. (2005). Ten usability heuristics. <https://www.nngroup.com/articles/ten-usability-heuristics/>. Last accessed 5 January 2022.
- [32] Nielsen, J. and Molich, R. (1990). Heuristic evaluation of user interfaces. In *Proceedings of the SIGCHI conference on Human factors in computing systems*, pages 249–256.
- [33] Norman, D. A. (1981). Categorization of action slips. *Psychological review*, 88(1):1.
- [34] O’Conaill, B. and Frohlich, D. (1995). Timespace in the workplace: Dealing with interruptions. In *Conference companion on Human factors in computing systems*, pages 262–263.
- [35] Przybylski, A. K., Murayama, K., DeHaan, C. R., and Gladwell, V. (2013). Motivational, emotional, and behavioral correlates of fear of missing out. *Computers in human behavior*, 29(4):1841–1848.

- [36] Quesenbery, W. (2001). What does usability mean: Looking beyond 'ease of use'. In *Annual conference-society for technical communication*, volume 48, pages 432–436. Citeseer.
- [37] Quesenbery, W. (2004). Balancing the 5es of usability. *Cutter IT Journal*, 17(2):4–11.
- [38] Reason, J. (1984). Absent-mindedness and cognitive control. *Everyday memory, actions and absentmindedness*, pages 113–132.
- [39] Rezaee, F. and Pedret, A. (2018). A critical review of silicon valley solutions for smartphone addiction. *Ideology Journal of Arts and Social Science*, 3(3):189–196.
- [40] Schmider, E., Ziegler, M., Danay, E., Beyer, L., and Bühner, M. (2010). Is it really robust? *Methodology*.
- [41] Schüll, N. D. (2012). *Addiction by design*. Princeton University Press.
- [42] Seligman, M. (2018). Perma and the building blocks of well-being. *The Journal of Positive Psychology*, 13(4):333–335.
- [43] Seligman, M. E. (2012). *Flourish: A visionary new understanding of happiness and well-being*. Simon and Schuster.
- [44] Skaugset, L. M., Farrell, S., Carney, M., Wolff, M., Santen, S. A., Perry, M., and Cico, S. J. (2016). Can you multitask? evidence and limitations of task switching and multitasking in emergency medicine. *Annals of emergency medicine*, 68(2):189–195.
- [45] Stringham, J. M., Stringham, N. T., and O'Brien, K. J. (2017). Macular carotenoid supplementation improves visual performance, sleep quality, and adverse physical symptoms in those with high screen time exposure. *Foods*, 6(7):47.
- [46] Vanden Abeele, M. M. (2021). Digital wellbeing as a dynamic construct. *Communication Theory*, 31(4):932–955.
- [47] Zhang, J., Patel, V. L., Johnson, T. R., and Shortliffe, E. H. (2004). A cognitive taxonomy of medical errors. *Journal of biomedical informatics*, 37(3):193–204.